

TANGENT & NORMAL

1.01 The slope of the tangent to the curve $(y - x^5)^2 = x(1+x^2)^2$ at the point (1,3) is. [JEE(Adv.)2014]

वक्र $(y - x^5)^2 = x(1+x^2)^2$ पर स्थित बिन्दु (1,3) पर स्पर्श रेखा का ढाल होगा। [JEE(Adv.)2014]

Ans. 8

Sol. $(y - x^5)^2 = x(1+x^2)^2$

$$2(y - x^5) \left(\frac{dy}{dx} - 5x^4 \right) = (1+x^2)^2 + 2x(1+x^2) \cdot 2x$$

at point (1, 3)

$$\therefore 2(3-1) \left(\frac{dy}{dx} - 5 \right) = 4 + 8$$

$$\frac{dy}{dx} - 5 = \frac{12}{4} = 3$$

$$\frac{dy}{dx} = 8$$

MONOTONOCITY

Paragraph for Question nos. 2.01 to 2.02

Let $f(x) = (1-x)^2 \sin^2 x + x^2$ for all $x \in \mathbb{R}$ and let $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right) f(t) dt$ for all $x \in (1, \infty)$.

माना $f(x) = (1-x)^2 \sin^2 x + x^2$, सभी $x \in \mathbb{R}$ के लिये तथा $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t \right) f(t) dt$ सभी $x \in (1, \infty)$

2.01 Which of the following is true? [JEE-2012]

- (A) g is increasing on $(1, \infty)$
- (B) g is decreasing on $(1, \infty)$
- (C) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$
- (D) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

निम्न में से कौनसा कथन सत्य है?

[JEE-2012]

- (A) g, $(1, \infty)$ में वर्धमान है
- (B) g, $(1, \infty)$ में ह्रसमान है
- (C) g, $(1, 2)$ में वर्धमान है तथा $(2, \infty)$ में ह्रसमान है
- (D) g, $(1, 2)$ में ह्रसमान है तथा $(2, \infty)$ में वर्धमान है

Ans. B

Sol. $f(x) = (1-x)^2 \sin^2 x + x^2$

$$P : f(x) + 2x = 2(1 + x^2)$$

$$\Rightarrow (1 - x)^2 \sin^2 x + x^2 + 2x = 2 + 2x^2$$

$$\Rightarrow (1 - x)^2 \sin^2 x - x^2 + 2x - 2 = 0$$

$$(1 - x)^2 \cos^2 x + 1 = 0$$

which is not possible.

$\therefore P$ is false.

$$Q : 2f(x) + 1 = 2x(1 + x)$$

$$2x^2 + 2(1 - x)^2 \sin^2 x + 1 = 2x^2 + 2x$$

$$2(1 - x)^2 \sin^2 x - 2x + 1 = 0.$$

$$\text{Let } h(x) = 2(1 - x)^2 \sin^2 x - 2x + 1,$$

$$\text{clearly } h(1) = -1$$

$$\text{and } h(x) = 2(x^2 - 2x + 1)\sin^2 x - 2x + 1$$

$$= x^2 \left[2 \left(1 - \frac{2}{x} + \frac{1}{x^2} \right) \cdot \sin^2 x - \frac{2}{x} + \frac{1}{x^2} \right]$$

$$\therefore h(x) \rightarrow \infty \text{ as } x \rightarrow \infty.$$

$\therefore$ By intermediate value theorem

$h(x) = 0$ has a root which is greater than 1.

Hence Q is true.

2.02 Consider the statements :

[JEE-2012]

P : There exists some $x \in \mathbb{R}$ such that $f(x) + 2x = 2(1 + x^2)$

Q : There exists some $x \in \mathbb{R}$ such that $2f(x) + 1 = 2x(1 + x)$

Then :

(A) both P and Q are true

(B) P is true and Q are false

(C) P is false and Q is true

(D) both P and Q are false

निम्न कथनों पर विचार कीजिये :

[JEE-2012]

P : किसी $x \in \mathbb{R}$ के लिये $f(x) + 2x = 2(1 + x^2)$ होगा

Q : किसी $x \in \mathbb{R}$ के लिये $2f(x) + 1 = 2x(1 + x)$ होगा

तो

(A) P तथा Q दोनों सत्य है

(B) P सत्य है तथा Q असत्य है

(C) P असत्य है तथा Q सत्य है

(D) P तथा Q दोनों असत्य है

Ans. C

Sol. $g(x) = \int_1^x \left(\frac{2(t-1)}{(t+1)} - \ln t \right) f(t) dt$

$$g'(x) = \left(\frac{2(x-1)}{x+1} - \ln x \right) f(x)$$

$$f(x) > 0 \quad \forall x \in \mathbb{R}$$

Suppose.

$$h(x) = \frac{2(x-1)}{x+1} - \ln x$$

$$h(x) = 2 - \left(\frac{4}{x+1} + \ln x \right)$$

$$h'(x) = \frac{4}{(x+1)^2} - \frac{1}{x}$$

$$h'(x) = -\frac{(x-1)^2}{x(x+1)^2}$$

$$h'(x) < 0$$

So $h(x)$ is decreasing

$$\text{so } h(x) < h(1) \quad \forall x > 1$$

$$h(x) < 0 \quad \forall x > 1$$

$$\text{So } g'(x) = h(x) f(x)$$

$$g'(x) < 0 \quad \forall x > 1$$

$g(x)$ is decreasing in $(1, \infty)$.

2.03 The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$, is :

[JEE(Adv.)2013]

$(-\infty, \infty)$ में बिन्दुओं की संख्या, जिनके लिए $x^2 - x \sin x - \cos x = 0$, है :

[JEE(Adv.)2013]

(A) 6

(B) 4

(C) 2

(D) 0

Ans. C

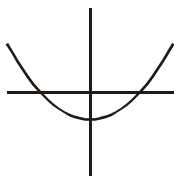
Sol. $f(x) = x^2 - x \sin x - \cos x$

$$f'(x) = 2x - x \cos x - \sin x + \sin x$$

$$= x(2 - \cos x)$$

$$\begin{array}{c} - \quad + \\ \hline 0 \end{array}$$

∴ graph of $f(x)$ will be



∴ $f(x)$ is zero for 2 values of x

∴ (c)

2.04 For every pair of continuous functions $f, g [0,1] \rightarrow \mathbb{R}$ such that $\max \{f(x): x \in [0,1]\} = \max \{g(x): x \in [0,1]\}$, the correct statement(s) is(are) : [JEE(Adv.)2014]

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
- (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
- (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0,1]$
- (D) $(f(c))^2 = (g(c))^2$ for some $c \in [0,1]$

प्रत्येक सतत फलन $f, g [0,1] \rightarrow \mathbb{R}$ के लिये $\max \{f(x): x \in [0,1]\} = \max \{g(x): x \in [0,1]\}$, सही कथन होंगे :

[JEE(Adv.)2014]

- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$, किसी $c \in [0,1]$ के लिए
- (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$, किसी $c \in [0,1]$ के लिए
- (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$, किसी $c \in [0,1]$ के लिए
- (D) $(f(c))^2 = (g(c))^2$, किसी $c \in [0,1]$ के लिए

Ans. AD

Sol. Consider

$$h(x) = f(x) - g(x) \quad \text{Assume} \quad a < b$$

$$h(a) = \lambda - g(a) > 0$$

$$h(b) = f(b) - \lambda < 0$$

else if $a > b$ $h(a) < 0$ and $h(b) > 0$.

By intermediate value theorem $\Rightarrow h(c) = 0$ (1)

$$(A) \quad (f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$$

$$(f(c) - g(c))(f(c) + g(c) + 3) = 0$$

So there exist a 'c' : $f(c) - g(c)$

from (1).

Hence A is correct.

$$(D) \quad \text{Similarly } (f(c))^2 = (g(c))^2$$

$$(f(c) - g(c))(f(c) + g(c)) = 0$$

$\Rightarrow$ (D) is correct.

B & C are wrong as by counter eg

If $f(x) = g(x) = \lambda \neq 0$, then

$B \rightarrow \lambda^2 + \lambda = \lambda^2 + 3\lambda$ is not possible.

$C \rightarrow \lambda^2 + 3\lambda = \lambda^2 + \lambda$ is not possible.

- 2.05 Let $f, g : [-1, 2] \rightarrow \mathbb{R}$ be continuous functions which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points $-1, 0$ and 2 be as given in the following table : [JEE(Adv.)2015]

	$x = -1$	$x = 0$	$x = 2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

In each of the intervals $(-1, 0)$ and $(0, 2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is (are)

- (A) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0, 2)$
 (B) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(-1, 0)$
 (C) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(0, 2)$
 (D) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0, 2)$

माना कि $f, g : [-1, 2] \rightarrow \mathbb{R}$ सतत् फलन हैं जो की अन्तराल $(-1, 2)$ में दो बार अवकलनीय (twice differentiable) है। माना कि f और g के मान, बिन्दुओं $-1, 0$ और 2 पर निम्न सारणी में दर्शाए गए हैं : [JEE(Adv.)2015]

	$x = -1$	$x = 0$	$x = 2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

यदि प्रत्येक अन्तराल $(-1, 0)$ और $(0, 2)$ में फलन $(f - 3g)''$ कभी भी शून्य का मान नहीं लेता है, तब सही कथन है (हैं)

- (A) $(-1, 0) \cup (0, 2)$ में, $f'(x) - 3g'(x) = 0$ के तीन ही हल (exactly three solutions) हैं
 (B) $(-1, 0)$ के अन्तराल में $f'(x) - 3g'(x) = 0$ का एक ही हल है।
 (C) $(0, 2)$ में, $f'(x) - 3g'(x) = 0$ का एक ही हल (exactly one solutions) है
 (D) $f'(x) - 3g'(x) = 0$ के अन्तराल $(-1, 0)$ में दो ही हल (exactly two solutions) हैं और $(0, 2)$ में दो ही हल हैं

Ans. BC

Sol. Let $h(x) = f(x) - 3g(x)$

$$\left. \begin{array}{l} h(-1) = 3 \\ h(0) = 3 \end{array} \right\} \Rightarrow h'(x) = 0 \text{ has atleast one root in } (-1, 0) \text{ and atleast one root in } (0, 2)$$

$$h(2) = 3$$

But since $h''(x) = 0$ has no root in $(-1, 0)$ & $(0, 2)$ therefore $h'(x) = 0$ has exactly 1 root in $(-1, 0)$ & exactly 1 root in $(0, 2)$

Paragraph for question nos. 2.06 to 2.08

By appropriately matching the information given in the three columns of the following table.

Let $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$

- Column 1 contains information about zeros of $f(x)$, $f'(x)$ and $f''(x)$.
- Column 2 contains information about the limiting behavior of $f(x)$, $f'(x)$ and $f''(x)$ at infinity.
- Column 3 contains information about the increasing/decreasing nature of $f(x)$, $f'(x)$

Column-1	Column-2	Column-3
(I) $f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x \rightarrow \infty} f(x) = 0$	(P) f is increasing in $(0, 1)$
(II) $f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
(III) $f'(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$	(R) f' is increasing in $(0, 1)$
(IV) $f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim_{x \rightarrow \infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)

नीचे दी गयी टेबल के तीन कॉलमों में उपलब्ध सूचना का उपयुक्त ढंग से सुमेल कर प्रश्नों 9 से 11 के उत्तर दीजिये।

माना कि $f(x) = x + \log_e x - x \log_e x$, $x \in (0, \infty)$ है।

- कॉलम 1 में $f(x)$, $f'(x)$ तथा $f''(x)$ के शून्यों की सूचना दी गयी है।
 - कॉलम 2 में $f(x)$, $f'(x)$ तथा $f''(x)$ के अनन्त की तरफ सीमा पर व्यवहार की सूचना दी गयी है।
 - कॉलम 3 में $f(x)$ तथा $f'(x)$ के वर्धमान/ह्रासमान होने की प्रकृति की सूचना दी गयी है।
- | | | |
|---|---|-------------------------------------|
| (I) $f(x) = 0$ किसी $x \in (1, e^2)$ के लिये | (i) $\lim_{x \rightarrow \infty} f(x) = 0$ | (P) f $(0, 1)$ में वर्धमान है |
| (II) $f'(x) = 0$ किसी $x \in (1, e)$ के लिये | (ii) $\lim_{x \rightarrow \infty} f(x) = -\infty$ | (Q) f (e, e^2) में ह्रासमान है |
| (III) $f'(x) = 0$ किसी $x \in (0, 1)$ के लिये | (iii) $\lim_{x \rightarrow \infty} f'(x) = -\infty$ | (R) f' $(0, 1)$ में वर्धमान है |
| (IV) $f''(x) = 0$ किसी $x \in (1, e)$ के लिये | (iv) $\lim_{x \rightarrow \infty} f''(x) = 0$ | (S) f' (e, e^2) में ह्रासमान है |

2.06 Which of the following options is the only CORRECT combination? [JEE(Adv.)2017]

निम्न में से कौनसा विकल्प केवल सही संयोजन है?

[JEE(Adv.)2017]

- (A) (II) (ii) (Q) (B) (III) (iii) (R) (C) (IV) (iv) (S) (D) (I) (i) (P)

Ans. A

Sol. $f(x) = x + \ln x - x \ln x$

$$f'(x) = \frac{1 - x \ln x}{x}$$

$$f''(x) = -\frac{(1+x)}{x^2} < 0$$

$$\Rightarrow f'(x) \text{ is } \downarrow \forall x \in (0, \infty)$$

$$\lim_{x \rightarrow 0} f'(x) = \infty$$

$$f'(x) = 1$$

$$f'(e) = \frac{1-e}{e} < 0$$

$$f'(x) < 0 \forall x > e$$

$$f(1) = 1 \quad f(e^2) = e^2 + 2 - 2e^2 = 2 - e^2 < 0$$

For column I (I) & (II) are true

For column II (ii), (iii) and (iv) are true

For column III P, Q, S are true

2.07 Which of the following options is the only CORRECT combination?

[JEE(Adv.)2017]

निम्न में से कौनसा विकल्प केवल सही संयोजन है?

[JEE(Adv.)2017]

(A) (III) (iv) (P) (B) (II) (iii) (S) (C) (I) (ii) (R) (D) (IV) (i) (S)

Ans. B

Sol. $f(x) = x + \ln x - x \ln x$

$$f'(x) = \frac{1 - x \ln x}{x}$$

$$f''(x) = -\frac{(1+x)}{x^2} < 0$$

$$\Rightarrow f'(x) \text{ is } \downarrow \forall x \in (0, \infty)$$

$$\lim_{x \rightarrow 0} f'(x) = \infty$$

$$f(x) = 1$$

$$f(e) = \frac{1-e}{e} < 0$$

$$f(x) < 0 \quad \forall x > e$$

$$f(1) = 1 \quad f(e^2) = e^2 + 2 - 2e^2 = 2 - e^2 < 0$$

For column I (I) & (II) are true

For column II (ii), (iii) and (iv) are true

For column III P, Q, S are true

2.08 Which of the following option is the only INCORRECT combination?

[JEE(Adv.)2017]

निम्न में से कौनसा विकल्प केवल गलत संयोजन है?

[JEE(Adv.)2017]

(A) (I) (iii) (P) (B) (II) (iii) (P) (C) (III) (i) (R) (D) (II) (iv) (Q)

Ans. C

Sol. $f(x) = x + \ln x - x \ln x$

$$f'(x) = \frac{1 - x \ln x}{x}$$

$$f''(x) = -\frac{(1+x)}{x^2} < 0$$

$\Rightarrow f'(x)$ is $\downarrow \forall x \in (0, \infty)$

$$\lim_{x \rightarrow 0} f'(x) = \infty$$

$$f'(x) = 1$$

$$f'(e) = \frac{1-e}{e} < 0$$

$$f'(x) < 0 \forall x > e$$

$$f(1) = 1 \quad f(e^2) = e^2 + 2 - 2e^2 = 2 - e^2 < 0$$

For column I (I) & (II) are true

For column II (ii), (iii) and (iv) are true

For column III P, Q, S are true

2.09 If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and $f\left(\frac{1}{2}\right) = \frac{1}{2}$, $f(1) = 1$, then :

[JEE(Adv.)2017]

यदि $f: \mathbb{R} \rightarrow \mathbb{R}$ एक इस प्रकार का द्विअवकलनीय फलन है, कि सभी $x \in \mathbb{R}$ के लिये $f''(x) > 0$ एवं $f\left(\frac{1}{2}\right) = \frac{1}{2}$, $f(1) = 1$ है, तब

: [JEE(Adv.)2017]

- (A) $\frac{1}{2} < f'(1) \leq 1$ (B) $0 < f'(1) \leq \frac{1}{2}$ (C) $f'(1) \leq 0$ (D) $f'(1) > 1$

Ans. D

Sol. $f''(x) > 0$ for all $x \in \mathbb{R}$, $f(1/2) = 1/2$, $f(1) = 1$
 $\Rightarrow f'(x)$ increases
 Let $g(x) = f(x) - x$, $x \in [1/2, 1)$
 $f'(x) = 1$ has atleast one real root in $(1/2, 1)$
 Hence $f'(x)$ increases $\Rightarrow f'(1) > 1$

2.10 For every twice differentiable function $f: \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE? [JEE(Adv.)2018]

- (A) There exist $r, s \in \mathbb{R}$, where $r < s$, such that f is one-one on the open interval (r, s)
 (B) There exists $x_0 \in (-4, 0)$ such that $|f'(x_0)| \leq 1$
 (C) $\lim_{x \rightarrow \infty} f(x) = 1$
 (D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \neq 0$

प्रत्येक द्विअवकलनीय फलन $f: \mathbb{R} \rightarrow [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, के लिये निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)? [JEE(Adv.)2018]

- (A) ऐसे $r, s \in \mathbb{R}$, जहाँ $r < s$, का अस्तित्व है जिनके लिये f खुले अन्तराल (r, s) पर एकैक है
 (B) ऐसे $x_0 \in (-4, 0)$ का अस्तित्व है जिसके लिये $|f'(x_0)| \leq 1$
 (C) $\lim_{x \rightarrow \infty} f(x) = 1$
 (D) ऐसे $\alpha \in (-4, 4)$ का अस्तित्व है जिसके लिये $f(\alpha) + f''(\alpha) = 0$ और $f'(\alpha) \neq 0$

Ans. ABD

Sol. $f^2(0) + (f'(0))^2 = 85$ $f: \mathbb{R} \rightarrow [-2, 2]$

(A) This is true for every continuous function

$$(b) \quad f'(c) = \frac{f(-4) - f(0)}{-4 - 0}$$

$$|f'(c)| = \left| \frac{f(-4) - f(0)}{4} \right|$$

$$-2 \leq f(-4) \leq 2$$

$$-2 \leq f(0) \leq 2$$

$$4 \leq f(-4) - f(0) \leq 4$$

This $|f'(c)| \leq 1$

$$(C) \quad \lim_{x \rightarrow \infty} f(x) = 1$$

Note $f(x)$ should have a bound ∞ which can be concluded by considering

$$f(x) = 2 \sin\left(\frac{\sqrt{85}x}{2}\right)$$

$$f(x) = \sqrt{85} \cos\left(\frac{\sqrt{85}x}{2}\right)$$

$$f(x) = \sqrt{85} \cos\left(\frac{\sqrt{85}x}{2}\right)$$

$$f^2(0) + (f'(0))^2 = 85$$

and $\lim_{x \rightarrow \infty} f(x)$ does not exist.

(D) Consider $H(x) = f^2(x) + (f'(x))^2$

$$H(0) = 85$$

By (B) choice there exists some x_0 such that $(f'(x_0))^2 \leq 1$ for some x_0 in $(-4, 0)$

$$\text{hence } H(x_0) = f^2(x_0) + (f'(x_0))^2 \leq 4 + 1$$

$$H(x_0) \leq 5$$

Hence let $p \in (-4, 0)$ for which $H(p) = 5$

(note that we have considered p as largest such negative number)

similarly let q be smallest positive number $\in (0, 4)$ such that $H(q) = 5$

Hence by Rolle's theorem is (p, q)

$H'(c) = 0$ for some $c \in (-4, 4)$ and since $H(x)$ is greater than 5 as we move from $x = p$ to $x = q$ and $f^2(x) \leq 4$
 $\Rightarrow (f'(x))^2 \geq 1$ in (p, q)

Thus $H'(c) = 0 \Rightarrow f'f + f'f' = 0$
 so $f + f' = 0$ and $f' \neq 0$

- 2.11 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$ Then which of the following statements is (are) TRUE? [JEE(Adv.)2021]
- (A) f is decreasing in the interval $(-2, -1)$ (B) f is increasing in the interval $(1, 2)$
- (C) f is onto (D) Range of f is $\left[-\frac{3}{2}, 2\right]$

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ निम्न प्रकार परिभाषित है :

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

तब निम्न में से कौनसा/कौनसे कथन सही है/हैं ?

- (A) f , अन्तराल $(-2, -1)$ में ह्रासमान है (B) f , अन्तराल $(1, 2)$ में वर्धमान है
- (C) f आच्छादक है (D) f का परिसर $\left[-\frac{3}{2}, 2\right]$ है

Ans. AB

Sol. $f: \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

$$f'(x) = \frac{(2x - 3)(x^2 + 2x + 4) - (2x + 2)(x^2 - 3x - 6)}{(x^2 + 2x + 4)^2}$$

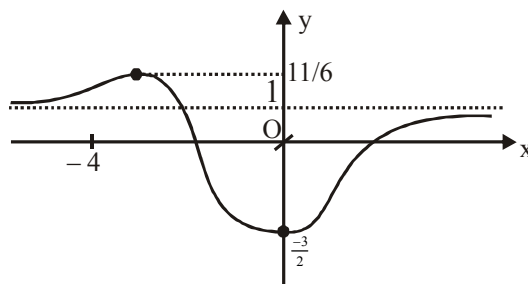
$$f'(x) = \frac{5x(x + 4)}{(x^2 + 2x + 4)^2} \Rightarrow \begin{array}{c} + \quad - \quad + \\ -4 \quad \quad 0 \end{array}$$

$f(x)$ is decreasing in $(-2, -1) \Rightarrow$ (A) is correct

$f(x)$ is increasing in $(1, 2) \Rightarrow$ (B) is correct

$$\text{Now, } f(-4) = \frac{16 + 12 - 6}{16 - 8 + 4} = \frac{11}{6}; f(0) = \frac{-3}{2}$$

$$\lim_{x \rightarrow \infty} f(x) = 1 \text{ \& } \lim_{x \rightarrow -\infty} f(x) = 1$$



Range is $\left[\frac{-3}{2}, \frac{11}{6}\right] \Rightarrow$ (D) is incorrect

Range $\neq$ Co-domain $\Rightarrow$ (C) is incorrect

- 2.12 Let S be the set of all twice differentiable functions f from $\mathbb{R}$ to $\mathbb{R}$ such that $\frac{d^2f}{dx^2}(x) > 0$ for all $x \in (-1, 1)$. For $f \in S$, let X_f be the number of points $x \in (-1, 1)$ for which $f(x) = x$. Then which of the following statements is (are) true? [JEE(Adv.)2023]

माना कि S उन दो बार अवकलनीय (twice differentiable) फलनों $f: \mathbb{R} \rightarrow \mathbb{R}$ का समुच्चय है जो इस प्रकार हैं कि सभी $x \in (-1, 1)$

के लिए $\frac{d^2f}{dx^2}(x) > 0$ है। फलन $f \in S$ के लिए, माना कि X_f उन बिन्दुओं $x \in (-1, 1)$ की संख्या है जिनके लिए $f(x) = x$ है। तब

निम्न में से कौनसा (से) कथन सत्य है (हैं)?

[JEE(Adv.)2023]

(A) There exists a function $f \in S$ such that $X_f = 0$

एक फलन $f \in S$ का अस्तित्व है जिसके लिए $X_f = 0$

(B) For every function $f \in S$, we have $X_f \leq 2$

सभी फलनों $f \in S$ के लिए $X_f \leq 2$ है

(C) There exists a function $f \in S$ such that $X_f = 2$

एक फलन $f \in S$ का अस्तित्व है जिसके लिए $X_f = 2$

(D) There does **NOT** exist any function f in such that $X_f = 1$

S में ऐसे किसी फलन f का अस्तित्व नहीं है जिसके लिए $X_f = 1$

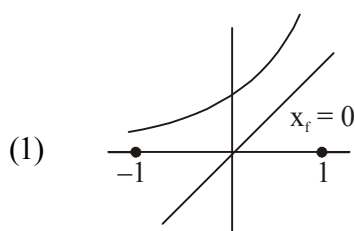
Ans. ABC

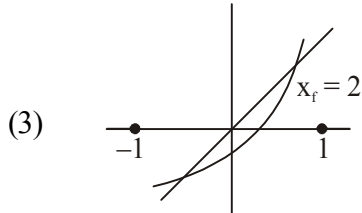
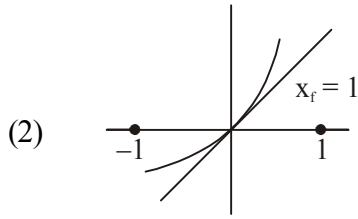
Sol. S = set of all twice differentiable functions $f: \mathbb{R} \rightarrow \mathbb{R}$

$$\frac{d^2f}{dx^2} > 0 \text{ in } (-1, 1)$$

Graph 'f' is Concave upward

Number of solutions of $f(x) = x \rightarrow x_f$





$\Rightarrow$ Graph of $y = f(x)$ can intersect graph of $y = x$ at atmost two points $\Rightarrow 0 \leq x_f \leq 2$

Aliter

$$\frac{d^2 f(x)}{dx^2} > 0$$

Let $\phi(x) = f(x) - x$

$$\phi''(x) > 0$$

$$\therefore \phi'(x) > 0 \text{ has atmost 1 root in } x \in (-1, 1)$$

$$\therefore \phi(x) > 0 \text{ has atmost 2 root in } x \in (-1, 1)$$

$$\therefore x_f \leq 2$$

2.13. Let the function $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \frac{\sin x (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)} + \frac{2 (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)}$

Then the number of solutions of $f(x) = 0$ in $\mathbb{R}$ is _____.

[JEE(Adv.)2024]

माना कि फलन (function) $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{\sin x (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)} + \frac{2 (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)}$ द्वारा

परिभाषित है। तब $\mathbb{R}$ में, $f(x) = 0$ के हलों (solutions) की संख्या _____ है।

[JEE(Adv.)2024]

Ans. 1

Sol. $f(x) = \frac{\sin x (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)} + \frac{2 (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)}$

$$\Rightarrow \frac{\sin x (x^{2023} + 2024x + 2025)}{e^{\pi x} (x^2 - x + 3)} [\sin x + 2] = 0$$

Now $\sin x + 2 = 0$

$\sin x = -2$ (No sol.)

So we need to solve

$$x^{2023} + 2024x + 2025 = 0$$

Let $g(x) = x^{2023} + 2024x + 2025$

Differentiate

$2023x^{2022} + 2024$ (Always positive)

So It cuts x-axis only once.

So exactly 1 solution.

MAXIMA & MINIMA

3.01 If $f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$ for all $x \in (0, \infty)$, then :

[JEE-2012]

- (A) f has a local maximum at $x = 2$ (B) f is decreasing on $(2, 3)$
 (C) there exists some $c \in (0, \infty)$ such that $f'(c) = 0$ (D) f has a local minimum at $x = 3$

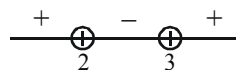
यदि $f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$, सभी $x \in (0, \infty)$ के लिए, तो :

[JEE-2012]

- (A) f का $x = 2$ पर स्थानीय उच्चिष्ठ है (B) f , $(2, 3)$ में ह्रासमान है
 (C) किसी $c \in (0, \infty)$ के लिए $f'(c) = 0$ होगा (D) f का $x = 3$ स्थानीय निम्निष्ठ है

Ans. ABCD

Sol. $f(x) = \int_0^x e^{t^2} (t-2)(t-3) dt$



$$\Rightarrow f'(x) = e^{x^2} (x-2)(x-3)$$

$$\therefore f'(2) = f'(3) = 0$$

$$\Rightarrow f''(c) = 0 \text{ for some } c \in (2, 3) \text{ (by Rolle's theorem)}$$

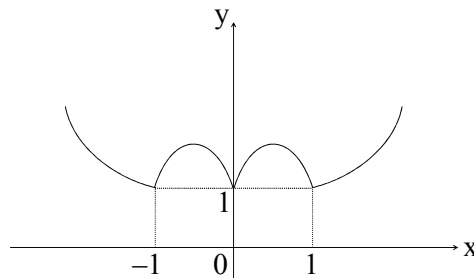
- 3.02 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is. [JEE-2012]

माना फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार है कि $f(x) = |x| + |x^2 - 1|$ तो f के स्थानीय निम्निष्ठ या स्थानीय उच्चिष्ठ वाले बिन्दुओं की संख्या होगी। [JEE-2012]

Ans. 5

Sol. $f(x) = |x| + |(x+1)(x-1)|$

$$\Rightarrow f(x) = \begin{cases} x^2 - x - 1 & x < -1 \\ -x^2 - x + 1 & -1 \leq x < 0 \\ -x^2 + x + 1 & 0 \leq x < 1 \\ x^2 + x - 1 & x \geq 1 \end{cases}$$



$\therefore f$ has 5 points where it attains either a local maximum or local minimum.

- 3.03 Let $p(x)$ be a real polynomial of least degree which has a local maximum at $x = 1$ and a local minimum at $x = 3$. If $p(1) = 6$ and $p(3) = 2$, then $p'(0)$ is. [JEE-2012]

माना कि $p(x)$ न्यूनतम घात का एक वास्तविक बहुपद है जिसका स्थानीय उच्चिष्ठ $x = 1$ पर तथा स्थानीय निम्निष्ठ $x = 3$ पर है। यदि $p(1) = 6$ तथा $p(3) = 2$, तो $p'(0)$ का मान होगा। [JEE-2012]

Ans. 9

- 3.04 The function $f(x) = 2|x| + |x+2| - ||x+2| - 2|x||$ has a local minimum or a local maximum at $x =$ [JEE(Adv.)2013]

फलन $f(x) = 2|x| + |x+2| - ||x+2| - 2|x||$ का एक स्थानीय न्यूनतम या एक स्थानीय अधिकतम जिन x के मान पर है, वह है : [JEE(Adv.)2013]

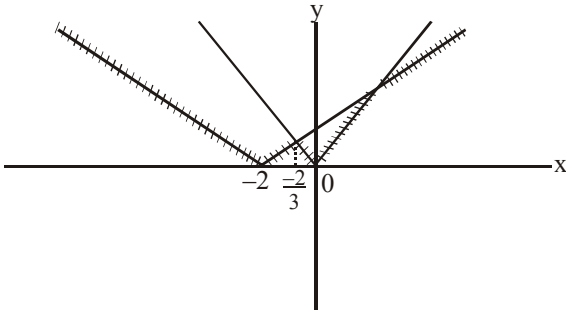
- (A) -2 (B) $-\frac{2}{3}$ (C) 2 (D) $\frac{2}{3}$

Ans. AB

Sol. $f(x) = (a+b) - |b-a|$

$$= \begin{cases} 2a, & a \leq b \\ 2b, & a > b \end{cases} = 2 \min(a, b)$$

where $a = 2|x|$, $b = |x + 2|$



$\therefore$ Local maxima and minima at $x = -2, -\frac{2}{3}$ & 0

Paragraph for question nos. 3.05 to 3.07

Let $f : (0, 1) \rightarrow \mathbb{R}$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0, 1]$.

माना कि $f : (0, 1) \rightarrow \mathbb{R}$ (सभी वास्तविक संख्याओं का समुच्चय) एक फलन है। मान लीजिये फलन f दो बार अवकलनीय है, $f(0) = f(1) = 0$ तथा $f''(x) - 2f'(x) + f(x) \geq e^x$, $x \in [0, 1]$ को सन्तुष्ट करता है।

3.05 Which of the following is true for $0 < x < 1$?

[JEE(Adv.)2013]

निम्न में से कौन $0 < x < 1$ के लिए सत्य है?

[JEE(Adv.)2013]

- (A) $0 < f(x) < \infty$ (B) $-\frac{1}{2} < f(x) < \frac{1}{2}$ (C) $-\frac{1}{4} < f(x) < 1$ (D) $-\infty < f(x) < 0$

Ans. D

Sol. $e^{-x}(f''(x) - 2f'(x) + f(x)) \geq 1$

$$D((f'(x) - f(x))e^{-x}) \geq 1$$

$$\Rightarrow D((f'(x) - f(x))e^{-x}) \geq 0$$

$$\Rightarrow (f'(x) - f(x))e^{-x} \text{ is an increasing function.}$$

As we know that $e^{-x}f(x)$ has local minima at $x = \frac{1}{4}$

$$e^{-x}(f'(x) - f(x)) = 0 \text{ at } x = \frac{1}{4}$$

$$\text{Let } F(x) = e^{-x}(f'(x) - f(x))$$

$$F(x) < 0 \text{ in } \left(0, \frac{1}{4}\right)$$

$$e^{-x}(f'(x) - f(x)) < 0 \text{ in } \left(0, \frac{1}{4}\right)$$

$$f'(x) < f(x) \text{ in } \left(0, \frac{1}{4}\right)$$

option C

3.06 If the function $e^{-x} f(x)$ assumes its minimum in the interval $[0, 1]$ at $x = \frac{1}{4}$, which of the following is true?

[JEE(Adv.)2013]

यदि फलन $e^{-x} f(x)$, अन्तराल $[0, 1]$ में अपना न्यूनतम मान $x = \frac{1}{4}$ पर लेता है, तब निम्न में से कौन सत्य है?

[JEE(Adv.)2013]

(A) $f(x) < f\left(\frac{1}{4}\right), \frac{1}{4} < x < \frac{3}{4}$

(B) $f(x) > f\left(\frac{1}{4}\right), 0 < x < \frac{1}{4}$

(C) $f(x) < f\left(\frac{1}{4}\right), 0 < x < \frac{1}{4}$

(D) $f(x) < f\left(\frac{1}{4}\right), \frac{3}{4} < x < 1$

Ans. C

Sol. $D(e^{-x}(f'(x) - f(x))) \geq 0 \forall x \in (0, 1)$

$$D(D(e^{-x}f(x))) \geq 0 \forall x \in (0, 1)$$

$$D^2(e^{-x}f(x)) \geq 0$$

$$\text{Let } F(x) = e^{-x}f(x)$$

$F''(x) > 0$ means it is concave upward.

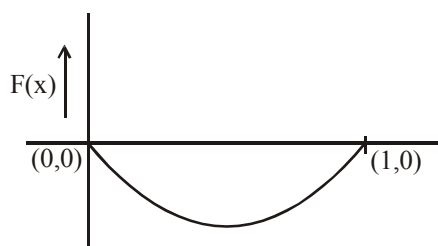
$$F(0) = F(1) = 0$$

$$F(x) < 0 \forall x \in (0, 1)$$

$$e^{-x}f(x) < 0 \forall x \in (0, 1)$$

$$f(x) < 0$$

Option D is possible



3.07 A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are :

[JEE(Adv.)2013]

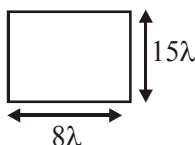
एक निश्चित परिमाप की आयताकार चादर को, जिसकी भुजाओं की लम्बाईयों 8 : 15 के अनुपात में हैं, सभी चारों किनारों से समान

क्षेत्रफल के वर्ग निकाल कर एक खुली आयताकार पेटी में परिवर्तित किया जाता है। यदि निकाले गये वर्गों का कुल क्षेत्रफल 100 है, तब परिणामी पेटी का आयतन महत्तम है। तब आयताकार चादर की भुजाओं की लम्बाइयाँ निम्न हैं : [JEE(Adv.)2013]

- (A) 24 (B) 32 (C) 45 (D) 60

Ans. AC

Sol. Where $P = 8\lambda + 15\lambda + 8\lambda + 15\lambda$ and λ is constant



Let removed length from each sides is x

Removed area is $4x^2 = 100 \Rightarrow x = 5$

$$V = (8\lambda - 2x)(15\lambda - 2x)x$$

$$V = 120\lambda^2 x - 46\lambda x^2 + 4x^3$$

$$\frac{dv}{dx} = 120\lambda^2 - 92\lambda x + 12x^2 = 0$$

$$\text{Put } x = 5 \Rightarrow 120\lambda^2 - 460\lambda + 300 = 0$$

$$12\lambda^2 - 46\lambda + 30 = 0$$

$$6\lambda^2 - 23\lambda + 15 = 0$$

$$(\lambda - 3)(6\lambda - 5) = 0$$

$$\lambda = 3 \text{ and } \lambda = \frac{5}{6}$$

$$\frac{d^2v}{dx^2} = -92\lambda + 24x = 120 - 92\lambda$$

$$\text{at } \lambda = 3 \Rightarrow \frac{d^2v}{dx^2} < 0$$

$$\text{at } \lambda = \frac{5}{6} \Rightarrow \frac{d^2v}{dx^2} > 0 \text{ (rejected)}$$

3.08 Let $a \in \mathbb{R}$ and let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^5 - 5x + a$. Then : [JEE(Adv.)2014]

- (A) $f(x)$ has three real roots if $a > 4$ (B) $f(x)$ has only one real root if $a > 4$
 (C) $f(x)$ has three real roots if $a < -4$ (D) $f(x)$ has three real roots if $-4 < a < 4$

माना कि $a \in \mathbb{R}$ तथा $f: \mathbb{R} \rightarrow \mathbb{R}$ निम्न के द्वारा $f(x) = x^5 - 5x + a$. परिभाषित है। तब : [JEE(Adv.)2014]

- (A) $a > 4$ के लिये $f(x)$ के तीन वास्तविक मूल (real roots) है
 (B) $a > 4$ के लिये $f(x)$ का केवल एक वास्तविक मूल है
 (C) $a < -4$ के लिये $f(x)$ के तीन वास्तविक मूल है

(D) $-4 < a < 4$ के लिये $f(x)$ के तीन वास्तविक मूल हैं

Ans. BD

Sol. $f(x) = x^5 - 5x + a = 0$

$$x^5 - 5x = -a$$

$$x(x^4 - 5) = -a$$

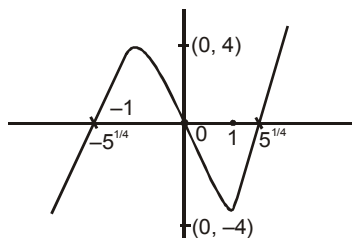
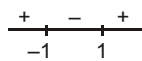
$$x(x^2 - \sqrt{5})(x^2 + \sqrt{5}) = -a$$

$$x(x - 5^{1/4})(x + 5^{1/4})(x^2 + \sqrt{5}) = -a \quad \dots(1)$$

$$f'(x) = 5x^4 - 5 = 0$$

$$(x^2 - 1)(x^2 + 1) = 0$$

$$(x - 1)(x + 1)(x^2 + 1) = 0$$



3.09 A cylindrical container is to be made from certain solid material with the following constraints. It has a fixed inner volume of $V \text{ mm}^3$, has a 2 mm thick wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container. If the volume of the material used to

make the container is minimum when the inner radius of the container is 10 mm, then the value of $\frac{V}{250\pi}$ is.

[JEE(Adv.)2015]

निम्न व्यवरोधों को संतुष्ट करते हुए एक बेलनाकार पात्र एक ठोस पदार्थ से बनाया है। पात्र का आन्तरिक आयतन V घन मिमी नियत है तथा इसकी दीवार 2 मिमी मोटाई की हैं एवं पात्र ऊपर से खुला है। पात्र का निचला तल 2 मिमी मोटाई वाला ठोस वृत्तीय डिस्क है तथा जिसकी त्रिज्या, पात्र की बाहरी त्रिज्या के बराबर है। यदि पात्र की आन्तरिक त्रिज्या 10 मिमी होने पर पदार्थ के न्यूनतम आयतन की

आवश्यकता होती हो, तब $\frac{V}{250\pi}$ का मान है।

[JEE(Adv.)2015]

Ans. 4

Sol. Volume of material $V = \pi r^2 h$

$$\Rightarrow V_1 = \pi(r + 2)^2 2 + \pi(r + 2)^2 h - \pi r^2 h \quad \Rightarrow V_1 = 2\pi(r + 2)^2 + \pi h(4 + 4r)$$

$$\Rightarrow V_1 = 2\pi(r + 2)^2 + 4\pi h(r + 1) \quad \Rightarrow V_1 = 2\pi \left((r + 2)^2 + \frac{2(r + 1)V}{\pi r^2} \right)$$

$$\Rightarrow \frac{dV_1}{dr} = 2\pi \left(2(r+2) + \frac{2v}{\pi} \left(\frac{-1}{r^2} - \frac{2}{r^3} \right) \right) = 0 \quad \Rightarrow \quad 24 + \frac{2V}{\pi} \left(\frac{-2-10}{10^3} \right) = 0$$

$$\Rightarrow \frac{24v}{10^3\pi} = 24 \quad \Rightarrow \quad v = 10^3\pi \quad \Rightarrow \quad \frac{v}{250\pi} = 4$$

3.10 Let $f : \mathbb{R} \rightarrow (0, \infty)$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable functions such that f' and g'' are continuous functions of

$\mathbb{R}$. Suppose $f'(2) = g(2) = 0$, $f''(2) \neq 0$ and $g'(2) \neq 0$. If $\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$, then. [JEE(Adv.)2016]

- (A) f has a local minimum at $x = 2$ (B) f has a local maximum at $x = 2$
 (C) $f''(2) > f(2)$ (D) $f(x) - f''(x) = 0$ for at least one $x \in \mathbb{R}$

माना कि $f : \mathbb{R} \rightarrow (0, \infty)$ और $g : \mathbb{R} \rightarrow \mathbb{R}$ ऐसे दो बार अवकलनीय फलन हैं कि $\mathbb{R}$ पर f'' और g'' संतत फलन हैं। मान लीजिये कि

$f'(2) = g(2) = 0$, $f''(2) \neq 0$ और $g'(2) \neq 0$ है। यदि $\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$ है, तब : [JEE(Adv.)2016]

- (A) $x = 2$ पर f का स्थानीय निम्नतम है (B) $x = 2$ पर f स्थानीय उच्चतम है
 (C) $f''(2) > f(2)$ (D) कम से कम एक $x \in \mathbb{R}$ के लिए $f(x) - f''(x) = 0$

Ans. AD

Sol. $\lim_{x \rightarrow 2} \frac{f(x)g(x)}{f'(x)g'(x)} = 1$

$$\lim_{x \rightarrow 2} \frac{f(x)g'(x) + f'(x)g(x)}{f'(x)g''(x) + f''(x)g'(x)} = 1 \quad \Rightarrow \quad \lim_{x \rightarrow 2} \frac{f(x)}{f''(x)} = 1$$

$$\Rightarrow f(2) = f''(2) = +ve \quad (\because \text{co-domain } (0, \infty))$$

Hence f has local maximum at $x = 2$.

3.11 If $f(x) = \begin{vmatrix} \cos(2x) & \cos(2x) & \sin(2x) \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$, then : [JEE(Adv.)2017]

- (A) $f(x)$ attains its maximum at $x = 0$ (B) $f(x)$ attains its minimum at $x = 0$
 (C) $f'(x) = 0$ at more than three points in $(-\pi, \pi)$ (D) $f'(x) = 0$ at exactly three points in $(-\pi, \pi)$

यदि $f(x) = \begin{vmatrix} \cos(2x) & \cos(2x) & \sin(2x) \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$, तब :

[JEE(Adv.)2017]

(A) $x = 0$ पर $f(x)$ का उच्चिष्ठ है

(B) $x = 0$ पर $f(x)$ का निम्निष्ठ है

(C) $(-\pi, \pi)$ में तीन से अधिक बिन्दुओं पर $f'(x) = 0$ है

(D) $(-\pi, \pi)$ में ठीक तीन बिन्दुओं पर $f'(x) = 0$ है

Ans. AC

Sol. $\begin{vmatrix} \cos 2x & \cos 2x & \sin 2x \\ -\cos x & \cos x & -\sin x \\ \sin x & \sin x & \cos x \end{vmatrix}$

$$= \cos 2x - \cos 2x(-\cos^2 x + \sin^2 x) + \sin 2x(-2\sin x \cos x)$$

$$f(x) = \cos 4x + \cos 2x$$

$$\therefore f(x) = 2\cos^2 2x + \cos 2x - 1$$

$$\text{Let } \cos 2x = t$$

$$\Rightarrow f(x) = 2t^2 + t - 1 \text{ and } t \in [-1, 1]$$

$$f(x) \text{ attains its minima at } t = -\frac{1}{4} \in [-1, 1]$$

$$f(x), t = -\frac{1}{4} \in [-1, 1]$$

$$\therefore f(x)|_{\min} = \frac{2}{16} - \frac{1}{4} - 1 = -\frac{9}{8}$$

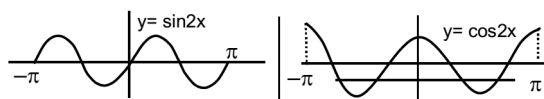
$$\therefore f(x)|_{\max} = 2 + 1 - 1 = 2 \dots \dots \dots (\text{when } \cos 2x = 1)$$

$$f'(x) = -4 \sin 4x - 2 \sin 2x$$

$$f'(x) = 0 \Rightarrow 4 \sin 4x + 2 \sin 2x = 0$$

$$\Rightarrow 8 \sin 2x \cos 2x + 2 \sin 2x = 0$$

$$\Rightarrow 2 \sin 2x(4 \cos 2x + 1) = 0 \Rightarrow \sin 2x = 0 \text{ or } \cos 2x = -\frac{1}{4}$$



3.12 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} x^5 + 5x^4 + 10x^3 + 10x^2 + 3x + 1, & x < 0; \\ x^2 - x + 1, & 0 \leq x < 1; \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3}, & 1 \leq x < 3; \\ (x-2)\log_e(x-2) - x + \frac{10}{3}, & x \geq 3 \end{cases}$$

Then which of the following options is/are correct :

[JEE(Adv.)2019]

- (A) f is increasing on $(-\infty, 0)$ (B) f' has a local maximum at $x = 1$
 (C) f' is NOT differentiable at $x = 1$ (D) f is onto

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}$ निम्न प्रकार से दिया है।

$$f(x) = \begin{cases} x^5 + 5x^4 + 10x^3 + 10x^2 + 3x + 1, & x < 0; \\ x^2 - x + 1, & 0 \leq x < 1; \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3}, & 1 \leq x < 3; \\ (x-2)\log_e(x-2) - x + \frac{10}{3}, & x \geq 3 \end{cases}$$

तब निम्न में से कौन सा (से) विकल्प सही है :

[JEE(Adv.)2019]

- (A) f अंतराल $(-\infty, 0)$ में वर्धमान है (B) f' का एक स्थानीय उच्चतम $x = 1$ पर है
 (C) $x = 1$ पर f' अवकलनीय नहीं है (D) f आच्छादक है

Ans. BCD

Sol. $f(x) = \begin{cases} (x+1)^5 - 2x & x < 0 \\ x^2 - x + 1 & 0 \leq x < 1 \\ \frac{2}{3}x^3 - 4x^2 + 7x - \frac{8}{3} & 1 \leq x < 3 \\ (x-2)\ln(x-2) - x + \frac{10}{3} & x \geq 3 \end{cases}$

$f(x)$ is continuous at $x = 0, x = 1, x = 3$

$\Rightarrow$ continuous for all $x \in \mathbb{R}$

$$f'(x) = \begin{cases} 5(x+1)^4 - 2 & x < 0 \\ 2x - 1 & 0 \leq x < 1 \\ 2x^3 - 8x + 7 & 1 < x < 3 \\ \ln(x-2) & x > 3 \end{cases}$$

$f'(x)$ is continuous at $x = 1$,

Discontinuous at $x = 0, 3$

Option (1) for $x < 0$ $f'(x)$ is changing its sign

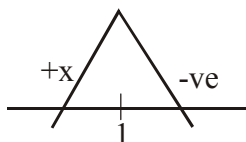
$\Rightarrow$ not increasing in $(0, \infty)$

Option (2) at $x = 1$

$$f''(x) = \begin{cases} 2 & 0 < x < 1 \\ 4x - 8 & 1 < x < 3 \end{cases}$$

$$f''(1^+) = -ve, f''(1^-) = +ve$$

f' has a local max^m at $x = 1$



Option (3) $f''(1^+) = -4 \neq f''(1^-) = 2$

f' is not differentiable at $x = 1$

Option (4) for $x \in \mathbb{R}$

for $x < 0$, $(x+1)^5 - 2x \in (-\infty, 1)$

for $x \geq 3$, $(x-2)\ln(x-2) - x + \frac{10}{3} \in \left[\frac{1}{3}, \infty\right)$

Range of $f(x) = (-\infty, \infty)$

3.13 Let $f(x) = \frac{\sin \pi x}{x^2}$, $x > 0$

Let $x_1 < x_2 < x_3 < \dots < x_n < \dots$ be all the points of local maximum of f and $y_1 < y_2 < y_3 < \dots < y_n < \dots$ be all the points of local minimum of f , then which of the following options is are correct : **[JEE(Adv.)2019]**

(A) $x_{n+1} - x_n > 2$ for every n

(B) $x_1 < y_1$

(C) $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ for every n

(D) $|x_n - y_n| > 1$ for every n

माना कि $f(x) = \frac{\sin \pi x}{x^2}, x > 0$

माना कि f के सभी स्थानीय उच्चतम (local maximum) बिन्दु $x_1 < x_2 < x_3 < \dots < x_n < \dots$ है और f के सभी स्थानीय न्यूनतम (local minimum) बिन्दु $y_1 < y_2 < y_3 < \dots < y_n < \dots$ हैं। तब निम्न में से कौन सा(से) विकल्प सही है (हैं)?

[JEE(Adv.)2019]

(A) प्रत्येक n के लिए $x_{n+1} - x_n > 2$ है

(B) $x_1 < y_1$

(C) प्रत्येक n के लिए $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ है

(D) प्रत्येक n के लिए $|x_n - y_n| > 1$ है

Ans. ACD

Sol. $f(x) = \frac{\sin \pi x}{x^2}, x > 0$

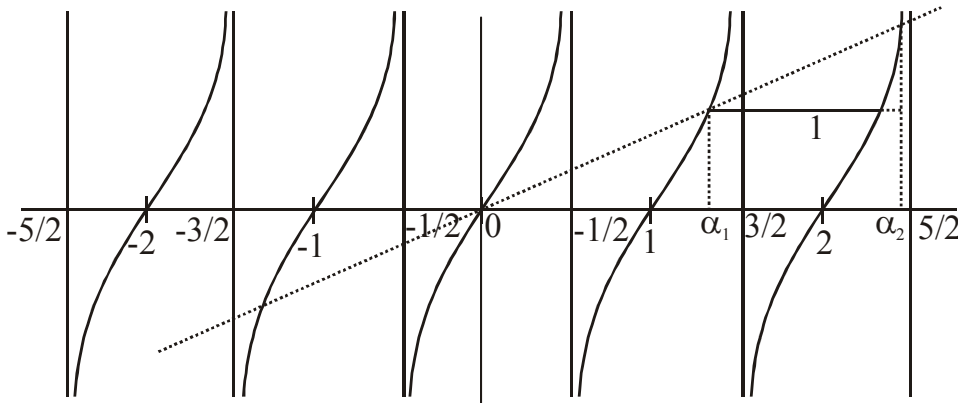
$$f'(x) = \frac{x^2 \cos(\pi x) \pi - \sin(\pi x) \cdot 2x}{x^4}$$

$$= \frac{[\pi x \cos(\pi x) - 2 \sin(\pi x)]}{x^3}$$

$$f'(x) = 0$$

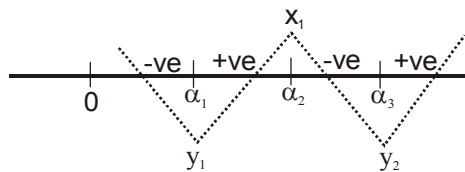
$$\Rightarrow \pi x \cos \pi x = 2 \sin \pi x$$

$$\tan \pi x = \frac{\pi x}{2}$$



At 0^+

$$f'(x) = -ve$$



$$\alpha_1 = x_1$$

$$\alpha_2 = y_1 \text{ (and so on)}$$

$$\alpha_3 = x_2$$

$$x_1 > y_1$$

$$x_1 \in (2, 5/2)$$

using periodicity

$$x_n \in (2n, 2n+1/2)$$

$$x_{n+1} - x_n > 2 \quad \forall n \text{ correct}$$

$$|x_n - y_n| > 1 \quad \forall n \text{ correct}$$

Question Stem for Question Nos. 3.14 and 3.15

Let $f_1 : (0, \infty) \rightarrow \mathbb{R}$ and $f_2 : (0, \infty) \rightarrow \mathbb{R}$ be defined by $f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j dt$, $x > 0$ and

$$f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \quad x > 0, \text{ where, for any positive integer } n \text{ and real numbers } a_1, a_2, \dots, a_n,$$

$\prod_{i=1}^n a_i$ denotes the product of $a_1, a_2, \dots, a_n$. Let m_i and n_i , respectively, denote the number of points of local minima and the number of points of local maxima of function f_i , $i = 1, 2$, in the interval $(0, \infty)$.

माना $f_1 : (0, \infty) \rightarrow \mathbb{R}$ तथा $f_2 : (0, \infty) \rightarrow \mathbb{R}$, $f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j dt$, $x > 0$ तथा

$$f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \quad x > 0, \text{ द्वारा परिभाषित है, जहाँ किसी धनात्मक पूर्णांक } n \text{ तथा वास्तविक संख्याओं}$$

$a_1, a_2, \dots, a_n$ के लिए $\prod_{i=1}^n a_i$ है, जो $a_1, a_2, \dots, a_n$ के गुणनफल को निरूपित करता है। माना m_i व n_i अंतराल $(0, \infty)$ में फलन f_i , $i = 1, 2$ के क्रमशः स्थानीय निम्निष्ठ बिन्दुओं तथा स्थानीय उच्चिष्ठ बिन्दुओं की संख्या को निरूपित करते हैं। तब

3.14 The value of $2m_1 + 3n_1 + m_1n_1$ is _____. [JEE(Adv.)2021]

$2m_1 + 3n_1 + m_1n_1$ का मान _____ है। [JEE(Adv.)2021]

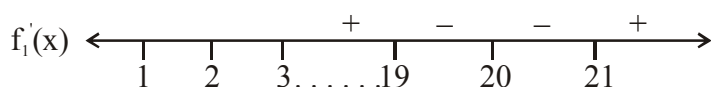
Ans. 57

3.15 The value of $6m_2 + 4n_2 + 8m_2n_2$ is _____. [JEE(Adv.)2021]

$6m_2 + 4n_2 + 8m_2n_2$ का मान _____ है। [JEE(Adv.)2021]

Ans. 6

Sol. $f_1'(x) = \prod_{j=1}^{21} (x-j)^j = (x-1)^1 (x-2)^2 (x-3)^3 \dots (x-21)^{21}$



Points of minima $\Rightarrow x = 21, 17, 13, 9, 5, 1 \Rightarrow m_1 = 6$

Points of maxima $\Rightarrow x = 19, 15, 11, 7, 3 \Rightarrow n_1 = 5$

$$2m_1 + 3n_1 + m_1n_1 = 57.$$

$$f_2'(x) = 98.50(x-1)^{49} - 600.49(x-1)^{48}$$

$$f_2'(x) = 4900(x-1)^{48}(x-7)$$

$$f_2'(x) \leftarrow \begin{array}{c} - \quad - \quad + \\ | \quad | \\ 1 \quad 7 \end{array} \rightarrow$$

Points of minima $\Rightarrow x = 7$

$$m_2 = 1$$

$$n_2 = 0$$

$$6m_2 + 4n_2 + 8m_2n_2 = 6.$$

3.16 Let $\alpha = \sum_{k=1}^{\infty} \sin^{2k} \left(\frac{\pi}{6} \right)$. Let $g: [0,1] \rightarrow R$ be the function defined by $g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$.

Then, which of the following statements is/are TRUE ?

[JEE(Adv.)2022]

(A) The minimum value of $g(x)$ is $2^{7/6}$

(B) The maximum value of $g(x)$ is $1 + 2^{1/3}$

(C) The function $g(x)$ attains its maximum at more than one point

(D) The function $g(x)$ attains its minimum at more than one point

माना कि $\alpha = \sum_{k=1}^{\infty} \sin^{2k} \left(\frac{\pi}{6} \right)$ है। माना कि फलन $g: [0,1] \rightarrow R$, $g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$ द्वारा परिभाषित है। तब निम्न कथनों

में से कौनसा सत्य है ?

[JEE(Adv.)2022]

(A) $g(x)$ का निम्नतम मान $2^{7/6}$ है

(B) $g(x)$ का उच्चतम मान $1 + 2^{1/3}$ है

(C) फलन $g(x)$ अपना उच्चतम मान एक से ज्यादा बिंदुओं पर प्राप्त करता है

(D) फलन $g(x)$ अपना निम्नतम मान एक से ज्यादा बिंदुओं पर प्राप्त करता है

Ans. (A, B, C)

Sol.
$$\alpha = \sum_{k=1}^{\infty} \left(\frac{1}{2} \right)^{2k} = \sum_{k=1}^{\infty} \left(\frac{1}{4} \right)^k = \frac{1/4}{1 - \frac{1}{4}} = \frac{1}{3}$$

Hence, $g(x) = 2^{x/3} + 2^{(1-x)/3}$

Now, $g'(x) = \frac{\ln 2}{3} \frac{(2^{2x/3} - 2^{1/3})}{2^{x/3}}$

$$g'(x) = 0 \text{ at } x = \frac{1}{2}$$

And, derivative changes sign from negative to positive at $x = \frac{1}{2}$, hence $x = \frac{1}{2}$ is point of local minimum as well as absolute minimum of $g(x)$ for $x \in [0, 1]$

$$\text{Hence, minimum value of } g(x) = g\left(\frac{1}{2}\right)$$

$$= 2^{1/6} + 2^{1/6} = 2^{7/6}$$

$\Rightarrow$ Option (A) is correct

Maximum value of $g(x)$ is either equal to $g(0)$ or $g(1)$.

$$g(0) = 1 + 2^{1/3}$$

$$g(1) = 2^{1/3} + 1$$

Hence (B) and (C) are also correct

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. The graphs $y = 2x^3 - 4x + 2$ and $y = x^3 + 2x - 1$ intersect in exactly 3 distinct points. The slope of the line passing through two of these points :

(A) is equal to 4 (B) is equal to 6 (C) is equal to 8 (D) is not unique

यदि वक्र $y = 2x^3 - 4x + 2$ और $y = x^3 + 2x - 1$, 3 बिन्दुओं पर प्रतिच्छेद करते हैं। तब इन बिन्दुओं में से दो से होकर जाने वाली रेखा की प्रवणता होगी :

(A) 4 (B) 6 (C) 8 (D) अद्वितीय नहीं

Ans. C

Sol. Let (x_1, y_1) and (x_2, y_2) are two of these points given $y = x^3 + 2x - 1$ and $y = 2x^3 - 4x + 2$

$$\therefore y_1 = 2x_1^3 - 4x_1 + 2 \quad \dots(1)$$

$$\text{and } 2y_1 = 2x_1^3 + 4x_1 - 2 \quad \dots(2)$$

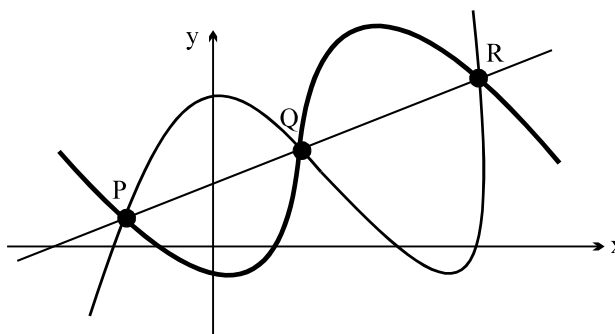
$$(2) - (1)$$

$$y_1 = 8x_1 - 4 \quad \dots(3)$$

$$\parallel y \quad y_2 = 8x_2 - 4 \quad \dots(4)$$

$$y_2 - y_1 = 8(x_2 - x_1)$$

$$\frac{y_2 - y_1}{x_2 - x_1} = 8$$



2. A function $y = f(x)$ has a second order derivative $f''(x) = 6(x - 1)$. If its graph passes through the point $(2, 1)$ and at that point the tangent to the graph is $y = 3x - 5$, then the function, is :

एक फलन $y = f(x)$ का दूसरे क्रम का अवकलज $f''(x) = 6(x - 1)$ है। यदि इसका आरेख बिन्दु $(2, 1)$ से होकर जाता है तथा वहाँ आरेख पर स्पर्शरेखा का समीकरण $y = 3x - 5$ है, तो फलन है :

(A) $(x - 1)^2$ (B) $(x - 1)^3$ (C) $(x + 1)^3$ (D) $(x + 1)^2$

Ans. B

Sol. $f''(x) = 6(x - 1)$

$$\Rightarrow f'(x) = 3x^2 - 6x + k \quad \underline{\hspace{2cm}} (1)$$

Slope of tangent at $(2, 1)$ is 3.

$$\Rightarrow 3 = 3 \times 2^2 - 6 \times 2 + k$$

$$\Rightarrow k = 3 \quad \underline{\hspace{2cm}} (2)$$

From (1) and (2)

$$f(x) = x^3 - 3x^2 + 3x + c$$

Using $f(x) = y = 1$ when $x = 2$, we get

$$1 = 2 + c \Rightarrow c = -1$$

Hence, the function is

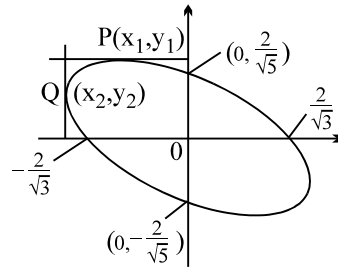
$$f(x) = x^3 - 3x^2 + 3x - 1$$

$$\Rightarrow f(x) = (x - 1)^3$$

3. The lines $y = -\frac{3}{2}x$ and $y = -\frac{2}{5}x$ intersect the curve $3x^2 + 4xy + 5y^2 - 4 = 0$ at the points P and Q respectively.

The tangents drawn to the curve at P and Q :

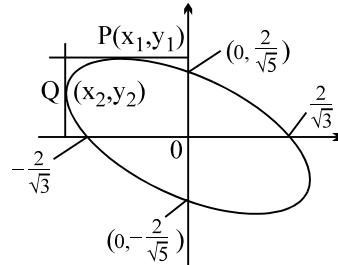
- (A) intersect each other at angle of 45°
 (B) are parallel to each other
 (C) are perpendicular to each other
 (D) none of these



रेखाएँ $y = -\frac{3}{2}x$ तथा $y = -\frac{2}{5}x$ वक्र $3x^2 + 4xy + 5y^2 - 4 = 0$ को बिन्दुओं P और Q पर क्रमशः प्रतिच्छेद करती है तब P और Q पर वक्र की स्पर्श रेखाएँ :

(A) एक दूसरे को 45° के कोण पर प्रतिच्छेद करती है

- (B) समान्तर है
 (C) लम्बवत् है
 (D) इनमें से कोई नहीं



Ans. C

Sol. $3x^2 + 4xy + 5y^2 - 4 = 0$

Differentiate

$$6x + 4xy' + 4y + 10yy' = 0$$

$$y'(4x + 10y) = -(6x + 4y)$$

$$y' = \frac{-(6x + 4y)}{4x + 10y}$$

Slope of the tangent at intersection by

$$y = -\frac{3}{2}x$$

$$y' = \frac{-\left(6x + 4\left(-\frac{3x}{2}\right)\right)}{4x + 10\left(-\frac{3x}{2}\right)}$$

$$y' = 0$$

Parallel to x-axis.

Slope of the tangent at intersection by

$$y = \frac{-2x}{5}$$

$$y' = \frac{-\left(6x + 4\left(-\frac{2x}{5}\right)\right)}{4x + 10\left(-\frac{2x}{5}\right)}$$

$$y' = \frac{-\left(6x + 4\left(-\frac{3x}{2}\right)\right)}{0}$$

Parallel to y-axis.

So both tangent are perpendicular.

4. The set of values of p for which the equation $|\ln x| - px = 0$ possess three distinct roots is :

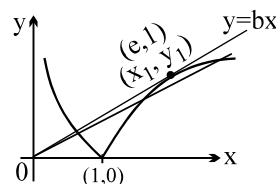
p का हल समुच्चय जबकि समीकरण $|\ln x| - px = 0$ के तीन भिन्न मूल हैं :

- (A) $\left(0, \frac{1}{e}\right)$ (B) $(0, 1)$ (C) $(1, e)$ (D) $(0, e)$

Ans. A

Sol. Hint: $1/x_1 = p = y_1/x_1 \Rightarrow y_1 = 1$ & $x_1 = e$.

Hence slope of $y = px$ i.e. $p \in (0, 1/e)$ for three roots



5. A curve is represented parametrically by the equations $x = t + e^{at}$ and $y = -t + e^{at}$ when $t \in \mathbb{R}$ and $a > 0$. If the curve touches the axis of x at the point A, then the coordinates of the point A are :

यदि किसी वक्र का प्राचल समीकरण $x = t + e^{at}$ तथा $y = -t + e^{at}$ जहाँ $t \in \mathbb{R}$ और $a > 0$ है। यदि वक्र x-अक्ष को बिन्दु A बिन्दु पर स्पर्श करता है तब A के निर्देशांक होंगे :

- (A) (1, 0) (B) (1/e, 0) (C) (e, 0) (D) (2e, 0)

Ans. D

Sol. $x = t + e^{at}$; $y = -t + e^{at}$

$$\frac{dx}{dt} = 1 + ae^{at}; \quad \frac{dy}{dt} = -1 + ae^{at}; \quad \frac{dy}{dx} = \frac{-1 + ae^{at}}{1 + ae^{at}}$$

at the point A, $y = 0$ and $\frac{dy}{dx} = 0$ for some $t = t_1$

$$\therefore ae^{at_1} = 1 \quad \dots(1);$$

$$\text{also } 0 = -t_1 + e^{at_1}; \therefore e^{at_1} = t_1 \quad \dots(2), \text{ putting this value in (1)}$$

$$\text{we get, at } t_1 = 1 \Rightarrow t_1 = \frac{1}{a}; \quad \text{now from (1) } ae = 1 \Rightarrow a = \frac{1}{e}$$

$$\text{hence } x_A = t_1 + e^{at_1} = e + e = 2e \Rightarrow A \equiv (2e, 0)$$

6. A horse runs along a circle with a speed of 20 km/hr. A lantern is at the centre of the circle. A fence is along the tangent to the circle at the point at which the horse starts. The speed with which the shadow of the horse moves along the fence at the moment when it covers $1/8$ of the circle in km/hr is :

एक घोड़ा एक वृत्त में 20 किमी/घण्टे की दर से दौड़ रहा है। एक लालटेन वृत्त के केन्द्र पर स्थित है। वृत्त के एक बिन्दु जहाँ से घोड़ा दौड़ना प्रारम्भ करता है, पर स्पर्श रेखा से एक चार दिवारी बनाई जाती है। उस दिवार पर घोड़े की परछाई की गति, जब घोड़ा वृत्त का $1/8$ तय कर चुका है, किमी/घण्टे में होनी चाहिये :

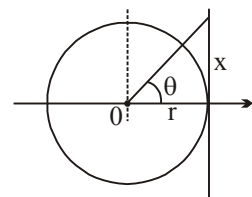
- (A) 20 (B) 40 (C) 30 (D) 60

Ans. B

Sol. $\tan \theta = x/r \Rightarrow x = r \tan \theta$

$$\Rightarrow dx/dt = r \sec^2 \theta (d\theta/dt) = r \omega \sec^2 \theta = v \sec^2 \theta$$

$$\text{where } \theta = 2\pi/8, dx/dt = v \sec^2(\pi/4) = 2v = 40 \text{ km/hr}; \quad \theta = 45^\circ$$



7. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \text{Min } \{f(t) : 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3 - x & ; 1 < x \leq 2 \end{cases}$ then $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has

the value equal to :

$$\text{यदि } f(x) = 4x^3 - x^2 - 2x + 1 \text{ तथा } g(x) = \begin{cases} \text{Min } \{f(t) : 0 \leq t \leq x\} & ; 0 \leq x \leq 1 \\ 3 - x & ; 1 < x \leq 2 \end{cases} \text{ तब } g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$$

का मान है :

(A) $\frac{7}{4}$

(B) $\frac{9}{4}$

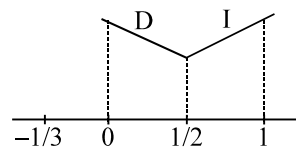
(C) $\frac{13}{4}$

(D) $\frac{5}{2}$

Ans. D

Sol. $f'(x) = 12x^2 - 2x - 2$
 $= 2[6x^2 - x - 1] = 2(3x + 1)(2x - 1)$

Hence $g(x) = \begin{cases} f(x) & \text{if } 0 \leq x < \frac{1}{2} \\ f\left(\frac{1}{2}\right) & \text{if } \frac{1}{2} \leq x \leq 1 \\ 3 - x & \text{if } 1 < x \leq 2 \end{cases}$



8. Let $f(x) = x^3 - 3x^2 + 2x$. If the equation $f(x) = k$ has exactly one positive and one negative solution then the value of k equals :

माना $f(x) = x^3 - 3x^2 + 2x$, यदि समीकरण $f(x) = k$ का एक ही धनात्मक और एक ही ऋणात्मक हल है तब $k =$

(A) $-\frac{2\sqrt{3}}{9}$

(B) $-\frac{2}{9}$

(C) $\frac{2}{3\sqrt{3}}$

(D) $\frac{1}{3\sqrt{3}}$

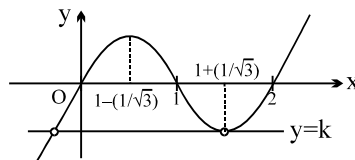
Ans. A

Sol. $f(x) = x(x^2 - 3x + 2)$

$f(x) = x(x-2)(x-1)$

graph of $y = f(x)$ is as shown

now $f(x) = k$ to have exactly one positive and negative solution



we have, $k = f\left(1 + \frac{1}{\sqrt{3}}\right)$ (think!) $1 - \frac{1}{\sqrt{3}}$ and $1 + \frac{1}{\sqrt{3}}$ are the roots of $f'(x) = 0$

$\therefore k = \underbrace{\left(1 + \frac{1}{\sqrt{3}}\right)}_x \underbrace{\left(\frac{1}{\sqrt{3}} - 1\right)}_{x-2} \underbrace{\left(\frac{1}{\sqrt{3}}\right)}_{x-1} = \left(\frac{1}{3} - 1\right)\left(\frac{1}{\sqrt{3}}\right) = \frac{-2}{3\sqrt{3}}$

9. Let $f(x)$ and $g(x)$ be two continuous functions defined from $\mathbb{R} \rightarrow \mathbb{R}$, such that $f(x_1) > f(x_2)$ and $g(x_1) < g(x_2)$, $\forall x_1 > x_2$, then solution set of $f(g(\alpha^2 - 2\alpha)) > f(g(3\alpha - 4))$ is :

माना $f(x)$ और $g(x)$, $\mathbb{R} \rightarrow \mathbb{R}$ पर परिभाषित दो सतत फलन है इस प्रकार की $f(x_1) > f(x_2)$ तथा

$g(x_1) < g(x_2)$, $\forall x_1 > x_2$, तब $f(g(\alpha^2 - 2\alpha)) > f(g(3\alpha - 4))$ का हल है :

(A) R

(B) ϕ

(C) (1, 4)

(D) $R - [1, 4]$ **Ans.** C**Sol.** obviously f is increasing and g is decreasing in (x_1, x_2) hence $f(g(\alpha^2 - 2\alpha)) > f(g(3\alpha - 4))$ as f is increasing

$$\Rightarrow g(\alpha^2 - 2\alpha) > g(3\alpha - 4)$$

 $\therefore \alpha^2 - 2\alpha < 3\alpha - 4$ as g is decreasing

$$\alpha^2 - 5\alpha + 4 < 0$$

$$(\alpha - 1)(\alpha - 4) < 0 \Rightarrow \alpha \in (1, 4)$$

10. Given $f'(1) = 1$ and $\frac{d}{dx}(f(2x)) = f'(x) \quad \forall x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals :

दिया है $f'(1) = 1$ तथा $\frac{d}{dx}(f(2x)) = f'(x) \quad \forall x > 0$. यदि $f'(x)$ अवकलनीय है तब $(2, 4)$ में किसी संख्या c का अस्तित्व इस प्रकार से होगा कि $f''(c) =$

(A) $-1/4$ (B) $-1/8$ (C) $1/4$ (D) $1/8$ **Ans.** B**Sol.** $f'(1) = 1; \quad 2 \cdot f'(2x) = f'(x)$

$$\text{put } x = 1, \quad f'(2) = \frac{f'(1)}{2} = \frac{1}{2}$$

$$\text{and } f'(4) = \frac{1}{2} f'(2) = \frac{1}{4}$$

applying LMVT for $y = f'(x)$ is $[2, 4]$

$$f''(c) = \frac{f'(4) - f'(2)}{2} = \frac{\frac{1}{4} - \frac{1}{2}}{2} = -\frac{1}{8}$$

11. If $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ where $0 < x < 1$, then :

(A) both ' f ' and ' g ' are increasing functions(B) ' f ' is decreasing and ' g ' is increasing function(C) ' f ' is increasing and ' g ' is decreasing function(D) both ' f ' and ' g ' are decreasing function

यदि $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ जहाँ $0 < x < 1$ हो, तो :

(A) 'f' एवं 'g' दोनों वर्द्धमान फलन हैं

(B) 'f' ह्रासमान फलन है तथा 'g' वर्द्धमान फलन है

(C) 'f' वर्द्धमान फलन है तथा 'g' ह्रासमान फलन है

(D) 'f' एवं 'g' दोनों ह्रासमान फलन हैं

Ans. C

Sol. $f(x) = \frac{x^2}{4} \cos \sec^2 \frac{x}{2}$

$$f'(x) = \frac{x}{2} \cos \sec^2 \frac{x}{2} \cot \frac{x}{2} \quad \left(\tan \frac{x}{2} - \frac{x}{2} \right)$$

For $0 < x < 1$, $\tan \frac{x}{2} > \frac{x}{2}$

$\Rightarrow f'(x) > 0 \Rightarrow f(x)$ is increasing.

12. Let $f(x) = \begin{cases} x^\alpha \ln x, & x > 0 \\ 0, & x = 0 \end{cases}$. Rolle's theorem is applicable to f for $x \in [0, 1]$, if $\alpha =$

माना $f(x) = \begin{cases} x^\alpha \ln x, & x > 0 \\ 0, & x = 0 \end{cases}$ फलन f के लिये $x \in [0, 1]$ में रोले प्रमेय सत्यापित है यदि $\alpha =$

(A) -2

(B) -1

(C) 0

(D) 1/2

Ans. D

Sol. Rolle's theorem is applicable to f for $x \in [0, 1]$.

This means that :

$f(1) = f(0)$ _____(1)

f is continuous in $x \in [0, 1]$ and f is differentiable for $x \in (0, 1)$ _____(2)

From (1) and (2) we get that : $f(1) = 0$

$0 = f(0) = \lim_{x \rightarrow 0} x^\alpha \ln x$ which is true only for positive values of α .

13. If f is a strictly increasing function, then $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ is equal to :

यदि f एक वर्द्धमान फलन है तो $\lim_{x \rightarrow 0} \frac{f(x^2) - f(x)}{f(x) - f(0)}$ का मान :

(A) 0

(B) 1

(C) -1

(D) 2

Ans. C

Sol.
$$\text{Limit} = \lim_{x \rightarrow 0} \left(\frac{f(x^2) - f(x)}{f(x) - 0} \right)$$
$$= \lim_{x \rightarrow 0} \left(\frac{f'(x^2) \cdot 2x - f'(x)}{f'(x)} \right)$$
$$= \lim_{x \rightarrow 0} \left(\frac{0 - f'(0)}{f'(0)} \right)$$
$$= -1$$

14. Given that $f(x)$ is continuously differentiable on $a \leq x \leq b$ where $a < b$, $f(a) < 0$ and $f(b) > 0$, which of the following are always true?

- (i) $f(x)$ is bounded on $a \leq x \leq b$.
 - (ii) The equation $f(x) = 0$ has at least one solution in $a < x < b$.
 - (iii) The maximum and minimum values of $f(x)$ on $a \leq x \leq b$ occur at points where $f'(c) = 0$.
 - (iv) There is at least one point c with $a < c < b$ where $f'(c) > 0$.
 - (v) There is at least one point d with $a < d < b$ where $f'(c) < 0$.
- (A) only (ii) and (iv) are true (B) all but (iii) are true
(C) all but (v) are true (D) only (i), (ii) and (iv) are true

दिया है फलन $f(x)$ अन्तराल $[a, b]$ में सतत् तथा अवकलनीय है जहाँ $a < b$, $f(a) < 0$ तथा $f(b) > 0$, तब निम्न में से कौन-कौन से हमेशा सत्य होंगे?

- (i) $f(x)$, $a \leq x \leq b$ में परिबद्ध है।
 - (ii) $f(x) = 0$ का कम से कम एक मूल $a < x < b$ में होगा।
 - (iii) $f(x)$ की अधिकतम और न्यूनतम मान $a \leq x \leq b$ में उन बिन्दुओं पर प्राप्त होगी जहाँ $f'(c) = 0$.
 - (iv) कम से कम एक बिन्दु c , $a < c < b$ होगा जहाँ $f'(c) > 0$
 - (v) कम से कम एक बिन्दु d , $a < d < b$ होगा जहाँ $f'(c) < 0$.
- (A) केवल (ii) और (iv) सत्य है (B) सभी लेकिन (iii) नहीं
(C) सभी लेकिन (v) नहीं (D) केवल (i), (ii) और (iv) सत्य

Ans. D

- Sol.**
- (i) This statement is true, every continuous function is bounded on a closed interval
 - (ii) True again, by Intermediate Value Theorem
 - (iii) Not true, because maximum and / or minimum values could also occur at a or b , without the derivatives being 0.

(iv) True. By the Mean Value Theorem there exist a point between a and b where the derivative is exactly $\frac{f(b) - f(a)}{b - a}$, a clearly positive value.

(v) Not always true, for example the function might be strictly increasing guaranteeing the derivative to be always positive.

Thus the true statements are (i), (ii) and (iv) and the correct answer is (D)

15. Give the correct order of initials **T** or **F** for following statements. Use **T** if statement is true and **F** if it is false.

Statement-1: If $f: \mathbb{R} \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$ is such that f is increasing in $(c - \delta, c)$ and f is decreasing in $(c, c + \delta)$ then f has a local maximum at c . Where δ is a sufficiently small positive quantity.

Statement-2 : Let $f: (a, b) \rightarrow \mathbb{R}$, $c \in (a, b)$. Then f can not have both a local maximum and a point of inflection at $x = c$.

Statement-3 : The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.

Statement-4 : Let $f: [c - 1, c + 1] \rightarrow [a, b]$ be bijective map such that f is differentiable at c then f^{-1} is also differentiable at $f(c)$.

निम्न कथनों के सत्य **T** और असत्य **F** होने का सही क्रम क्या है?

कथन-1: यदि $f: \mathbb{R} \rightarrow \mathbb{R}$ और $c \in \mathbb{R}$ इस प्रकार से है कि f अन्तराल $(c - \delta, c)$ में वर्धमान और f अन्तराल $(c, c + \delta)$ में ह्रसमान है तब f, c पर स्थानीय उच्चिष्ठ है जहाँ δ छोटी से छोटी धनात्मक संख्या है।

कथन-2 : माना $f: (a, b) \rightarrow \mathbb{R}$, $c \in (a, b)$ तब $f, x = c$ पर स्थानीय उच्चिष्ठ और नति परिवर्तन बिन्दु एक साथ नहीं देगा।

कथन-3 : फलन $f(x) = x^2 |x|$, $x = 0$ पर दो बार अवकलनीय है।

कथन-4 : माना $f: [c - 1, c + 1] \rightarrow [a, b]$ एकैकी आच्छादक फलन इसप्रकार है कि $f, x = c$ पर अवकलनीय है तब f^{-1} भी $f(c)$ पर अवकलनीय होगा।

(A) FFTF

(B) TTFT

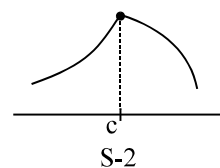
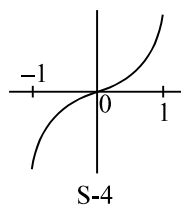
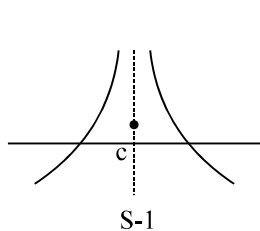
(C) FTTF

(D) TTTF

Ans. C

Sol. **S-1 :** f must be continuous $\Rightarrow$ false

S-2: $x=c$ is local max. but not inflection point $\Rightarrow$ True



S-4 : Consider $f(x) = x^3$ at $x = 0$

$f^{-1}(x) = x^{1/3}$, not differentiable at $x = 0$

16. In a regular triangular prism the distance from the centre of one base to one of the vertices of the other base is l . The altitude of the prism for which the volume is greatest :

एक समबाहु प्रिज्म में आधार के केन्द्र से दूसरे आधार के शीर्ष की दूरी l है। प्रिज्म की ऊँचाई ज्ञात कीजिए, जिस के लिए आयतन अधिकतम हो

(A) $\frac{l}{2}$

(B) $\frac{l}{\sqrt{3}}$

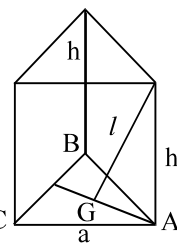
(C) $\frac{l}{3}$

(D) $\frac{l}{4}$

Ans. B

Sol. $AG = \frac{2}{3} \cdot \frac{a\sqrt{3}}{2} = \frac{a}{\sqrt{3}}$. Now $l^2 = \frac{a^2}{3} + h^2$; area (A) = $\frac{\sqrt{3}}{4} a^2$

$V(h) = \frac{\sqrt{3}}{4} a^2 h = \frac{\sqrt{3}}{4} h(l^2 - h^2)3$; $V'(h) = 0 \Rightarrow h = \frac{l}{\sqrt{3}}$; $V_{\max} = \frac{l^3}{2}$



17. Let h be a twice continuously differentiable positive function on an open interval J . Let $g(x) = \ln(h(x))$ for each $x \in J$

Suppose $(h'(x))^2 > h''(x)h(x)$ for each $x \in J$. Then :

(A) g is increasing on J

(B) g is decreasing on J

(C) g is concave up on J

(D) g is concave down on J

यदि h खुले अन्तराल J में दो बार अवकलनीय धनात्मक फलन है

माना $g(x) = \ln(h(x)) \quad \forall x \in J$

एवं $(h'(x))^2 > h''(x)h(x) \quad \forall x \in J$ तब :

(A) g, J में वर्धमान है

(B) g, J में ह्रासमान है

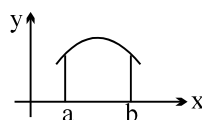
(C) g, J में ऊपर की ओर अवतल है

(D) g, J में नीचे की ओर अवतल है

Ans. D

Sol. Given $g(x) = \ln(h(x))$

$$g'(x) = \frac{h'(x)}{h(x)}$$



$$g''(x) = \frac{h(x)h''(x) - (h'(x))^2}{h^2(x)} < 0 \quad (\text{given})$$

$$\therefore g''(x) < 0 \Rightarrow g(x) \text{ is concave down}$$

18. If the function $f(x) = \frac{t+3x-x^2}{x-4}$, where ' t ' is a parameter has a minimum and a maximum then the range of values of ' t ' is :

यदि फलन $f(x) = \frac{t+3x-x^2}{x-4}$, जहाँ ' t ' प्राचल है का एक न्यूनतम और अधिकतम मान प्राप्त होता है तब ' t ' के मानों का परिसर

होगा:

(A) (0, 4)

(B) (0, ∞)

(C) (−∞, 4)

(D) (4, ∞)

Ans. C

Sol. $f(x) = \frac{t+3x-x^2}{x-4}$; $f'(x) = \frac{(x-4)(3-2x)-(t+3x-x^2)}{(x-4)^2}$

for maximum or minimum, $f'(x) = 0$

$$-2x^2 + 11x - 12 - t - 3x + x^2 = 0$$

$$-x^2 + 8x - (12 + t) = 0$$

for one M and m,

$$D > 0$$

$$64 - 4(12 + t) > 0$$

$$16 - 12 - t > 0 \Rightarrow 4 > t \quad \text{or} \quad t < 4$$

19. P and Q are two points on a circle of centre C and radius α , the angle PCQ being 2θ then the radius of the circle inscribed in the triangle CPQ is maximum when :

(A) $\sin \theta = \frac{\sqrt{3}-1}{2\sqrt{2}}$

(B) $\sin \theta = \frac{\sqrt{5}-1}{2}$

(C) $\sin \theta = \frac{\sqrt{5}+1}{2}$

(D) $\sin \theta = \frac{\sqrt{5}-1}{4}$

Ans. B

Sol. $r = \frac{\Delta}{s}$ where Δ = Area of triangle CPQ and s = semiperimeter of $\triangle CPQ$.

$$r = \frac{\alpha^2 \sin 2\theta}{2s} = \frac{\alpha^2 \sin 2\theta}{2\alpha + 2\alpha \sin \theta} = \frac{\alpha}{2} \cdot \frac{\sin 2\theta}{1 + \sin \theta}$$

Consider $f(\theta) = \frac{\sin 2\theta}{1 + \sin \theta}$

$$f'(\theta) = \frac{(1 + \sin \theta) 2 \cos 2\theta - \sin 2\theta \cdot \cos \theta}{(1 + \sin \theta)^2} = 0$$

$$2(1 + \sin \theta)(1 - 2\sin^2 \theta) - 2\sin \theta (1 - \sin^2 \theta) = 0$$

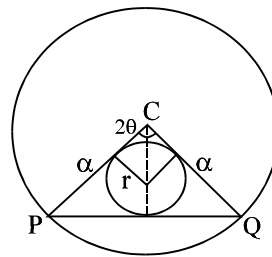
$$2(1 - 2\sin^2 \theta) = 2\sin \theta (1 - \sin \theta)$$

$$1 - 2\sin^2 \theta = \sin \theta - \sin^2 \theta$$

$$\sin^2 \theta + \sin \theta - 1 = 0$$

$$\sin \theta = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$\therefore \sin \theta = \frac{\sqrt{5}-1}{2}$$



20. The range of values of m for which the line $y = mx$ and the curve $y = \frac{x}{x^2 + 1}$ enclose a region, is :

m के मानों का परिसर क्या होगा जबकि $y = mx$ और वक्र $y = \frac{x}{x^2 + 1}$ एक परिबद्ध क्षेत्र बनाते हैं :

- (A) $(-1, 1)$ (B) $(0, 1)$ (C) $[0, 1]$ (D) $(1, \infty)$

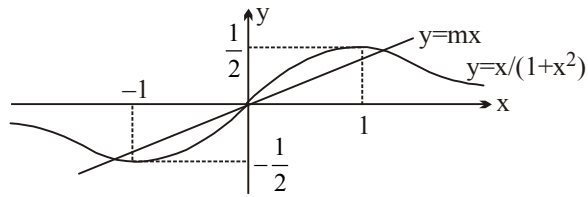
Ans. B

Sol. Solving

$$mx = \frac{x}{x^2 + 1} \Rightarrow x^2 + 1 = \frac{1}{m} \text{ or } x = 0$$

$$x^2 = \frac{1}{m} - 1 > 0 \text{ for a region}$$

$$\frac{m-1}{m} < 0 \Rightarrow m \in (0, 1)$$



Note: for $m = 0$ or 1 the line does not enclose a region

21. For a steamer the consumption of petrol (per hour) varies as the cube of its speed (in km). If the speed of the current is steady at C km/hr then the most economical speed of the steamer going against the current will be :

किसी पानी के जहाज के लिए एक घण्टे में ईंधन की खपत उसकी किलोमीटर में चाल के घन के समानुपाती है। यदि धारा की नियत चाल C km/hr है तो जहाज की सबसे कम खर्चीली गति धारा के विपरीत दिशा में क्या होनी चाहिये :

- (A) $1.25 C$ (B) $1.5 C$ (C) $1.75 C$ (D) $2 C$

Ans. B

Sol. Time of journey = $\frac{d}{V-C}$ when V is the speed in still water.

$$\therefore \text{petrol burnt per hour} = kV^3$$

$$\therefore \text{fuel cost} = \frac{kV^3 d}{V-C} = kd \frac{V^3}{V-C} = f(v)$$

now proceed

22. Let f, g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = x e^{x^2} + e^{-x^2}$ and $h(x) = x^2 e^{x^2} + e^{-x^2}$. If a, b and c denote respectively, the absolute maximum of f, g and h on $[0, 1]$, then :
- (A) $a = b$ and $c \neq b$ (B) $a = c$ and $a \neq b$ (C) $a \neq b$ and $c \neq b$ (D) $a = b = c$

माना f, g तथा h अन्तराल $[0, 1]$ में $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ तथा $h(x) = x^2e^{x^2} + e^{-x^2}$ द्वारा परिभाषित फलन है। यदि a, b तथा c, f, g तथा h के अन्तराल $[0, 1]$ पर परम उच्चिष्ठ मान है, तो :

- (A) $a = b$ तथा $c \neq b$ (B) $a = c$ तथा $a \neq b$ (C) $a \neq b$ तथा $c \neq b$ (D) $a = b = c$

Ans. D

Sol. If $x \in [0, 1]$

then $x^2 \leq x \leq 1$

$$x^2e^{x^2} \leq xe^{x^2} \leq e^{x^2}$$

Add e^{-x^2} to all sides

$$x^2e^{x^2} + e^{-x^2} \leq xe^{x^2} + e^{-x^2} \leq e^{x^2} + e^{-x^2}$$

$$\Rightarrow h(x) \leq g(x) \leq f(x) \quad \dots\dots\dots (i)$$

where, $f(x) = e^{x^2} + e^{-x^2}$

$$f'(x) = 2x(e^{x^2} - e^{-x^2}) > 0$$

$\Rightarrow f(x)$ has a maxima at $x = 1$

$$\Rightarrow a = e + \frac{1}{e}$$

$$h(x) = x^2e^{x^2} + e^{-x^2}$$

$$h'(x) = 2x^3e^{x^2} + 2xe^{x^2} - 2xe^{-x^2}$$

$$= 2x^3e^{x^2} + 2x(e^{x^2} - e^{-x^2}) > 0$$

$\Rightarrow h(x)$ has a maxima at $x = 1$

$$\Rightarrow c = e + \frac{1}{e}$$

$$\therefore h(x) \leq g(x) \leq f(x)$$

$\Rightarrow g(x)$ also has a maximum value at $x = 1$

$$\Rightarrow a = b = c$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

23. Consider the curve $f(x) = x^{1/3}$, then :

- (A) the equation of tangent at $(0, 0)$ is $x = 0$ (B) the equation of normal at $(0, 0)$ is $y = 0$
 (C) normal to the curve does not exist at $(0, 0)$ (D) $f(x)$ and its inverse meet at exactly 3 points

माना वक्र $f(x) = x^{1/3}$ है, तो :

- (A) $(0, 0)$ पर स्पर्श रेखा का समीकरण $x = 0$ है (B) $(0, 0)$ पर अभिलम्ब का समीकरण $y = 0$ है
 (C) $(0, 0)$ पर अभिलम्ब अस्तित्व में नहीं है (D) $f(x)$ तथा इसका प्रतिलोम ठीक तीन बिन्दुओं पर मिलते हैं

Ans. ABD

Sol. $f(x) = \frac{1}{3x^{2/3}}$

$f(0) \rightarrow \infty$ tangent is vertical at $x = 0$

Equation of tangent at $(0, 0)$ is $x = 0$

Equation of normal is $y = 0$

$f(x) = f^{-1}(x)$

$x^{\frac{1}{3}} = x^3 \Rightarrow x^9 = x$

$\Rightarrow x = 0 ; 1 ; -1$

24. If the line, $ax + by + c = 0$ is a normal to the curve $xy = 2$, then :

यदि रेखा $ax + by + c = 0$ वक्र $xy = 2$ पर अभिलम्ब हो, तो :

(A) $a < 0, b > 0$ (B) $a > 0, b < 0$ (C) $a > 0, b > 0$ (D) $a < 0, b < 0$

Ans. AB

Sol. Given $xy = 2 \Rightarrow y = \frac{2}{x}$

$\Rightarrow \frac{dy}{dx} = -\frac{2}{x^2}$ slope of tangent

Thus, slope of normal = $\frac{x^2}{2}$ (which is always positive) and it is given that

$ax + by + c = 0$ is normal whose slope = $-\frac{a}{b}$.

$\Rightarrow -\frac{a}{b} > 0$ or $\frac{a}{b} < 0$

$\therefore a$ and b are of opposite sign.

Hence if $a > 0, b < 0$ or $a < 0, b > 0$

25. The normal to the curve $x = a(\cos \theta + \theta \sin \theta)$, $y = a(\sin \theta - \theta \cos \theta)$ at any point ' θ ' is such that :

(A) it passes through the origin (B) it makes an angle $\frac{\pi}{2} + \theta$ with the x-axis

(C) it passes through $\left(a \frac{\pi}{2}, -a\right)$ (D) it is at a constant distance from the origin

वक्र $x = a(\cos \theta + \theta \sin \theta)$, $y = a(\sin \theta - \theta \cos \theta)$ के बिन्दु ' θ ' पर खींचा गया अभिलम्ब ऐसा है, कि :

- (A) वह मूल बिन्दु से होकर जाता है
(B) वह x-अक्ष के साथ $\frac{\pi}{2} + \theta$ का कोण बनाता है
(C) वह बिन्दु $\left(a\frac{\pi}{2}, -a\right)$ से होकर जाता है
(D) वह मूल बिन्दु से अचर दूरी पर है

Ans. BD

Sol. $x = a(\cos \theta + \theta \sin \theta)$ & $y = a(\sin \theta - \theta \cos \theta)$

$$\frac{dx}{d\theta} = a(-\sin \theta + \sin \theta + \theta \cos \theta),$$

$$\frac{dy}{d\theta} = a(\cos \theta - \cos \theta + \theta \sin \theta)$$

$$\therefore \frac{dy}{dx} = \frac{\sin \theta}{\cos \theta}$$

$$\text{slope of normal} = -\frac{\cos \theta}{\sin \theta} = -\cot \theta$$

it makes angle $\left(\frac{\pi}{2} + \theta\right)$ with the x-axis

$$\text{eq of normal } y - a \sin \theta + a \theta \cos \theta = -\frac{\cos \theta}{\sin \theta}$$

$$(x - a \cos \theta - a \theta \sin \theta)$$

$$\Rightarrow x \cos \theta + y \sin \theta = a.$$

Hence it is at a constant distance 'a' from the origin.

26. Which of the following functions do not satisfy conditions of Rolle's Theorem?

निम्नलिखित में से कौनसे फलन रोल प्रमेय के प्रतिबन्धों को संतुष्ट नहीं करते हैं?

- (A) $e^x \sin x$, $x \in \left[0, \frac{\pi}{2}\right]$
(B) $(x+1)^2(2x-3)^5$, $x \in \left[-1, \frac{3}{2}\right]$
(C) $\sin |x|$, $x \in [\pi, 2\pi]$
(D) $\sin \frac{1}{x}$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Ans. AD

Sol. (A) $f(0) = 0$

$$f\left(\frac{\pi}{2}\right) = e^{\frac{\pi}{2}}$$

$$f(0) \neq f\left(\frac{\pi}{2}\right)$$

$$(D) f\left(\frac{-\pi}{2}\right) = -\sin\left(\frac{2}{\pi}\right)$$

$$f\left(\frac{\pi}{2}\right) = \sin\left(\frac{2}{\pi}\right)$$

$$f\left(-\frac{\pi}{2}\right) \neq f\left(\frac{\pi}{2}\right)$$

27. The function $y = 2x^2 - \ln|x|$ is monotonically increasing in the interval I_1 and monotonically decreasing in the interval I_2 , $x (\neq 0)$, then :

फलन $y = 2x^2 - \ln|x|$ अन्तराल I_1 में एकदिष्ट वर्द्धमान है तथा अन्तराल I_2 में एकदिष्ट ह्रासमान है $x (\neq 0)$, तो :

$$(A) I_1 = \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$$

$$(B) I_2 = \left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$$

$$(C) I_1 = \left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$$

$$(D) I_2 = \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$$

Ans. AB

Sol. $y = 2x^2 - \ln|x|$

$$\frac{dy}{dx} = 4x - \frac{1}{x} = \frac{(2x+1)(2x-1)}{x}$$

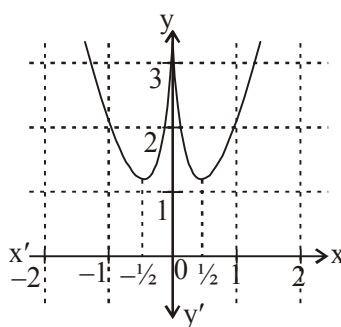
Critical points are 0, $1/2$, $-1/2$

Sign scheme of $\frac{dy}{dx}$ is

Clearly $f(x)$ increases in

$$\left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right) \text{ and } f(x) \text{ decreases}$$

$$\text{in } \left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right).$$



28. For the function $f(x) = x^4 (12 \ln x - 7)$:
- (A) the point $(1, -7)$ is the point of inflection (B) $x = e^{1/3}$ is the point of minima
 (C) the graph is concave downwards in $(0, 1)$ (D) the graph is concave upwards in $(1, \infty)$

फलन $f(x) = x^4 (12 \ln x - 7)$ के लिए :

- (A) बिन्दु $(1, -7)$ नति परिवर्तन बिन्दु है (B) $x = e^{1/3}$ निम्ननिष्ठि का बिन्दु है
 (C) आलेख $(0, 1)$ में उत्तल है (D) आरेख $(1, \infty)$ में अवतल है

Ans. ABCD

Sol. $y = f(x) = x^4 (12 \log_e x - 7), x > 0$

$$\therefore \frac{dy}{dx} = 16x^3 (3 \log_e x - 1) \text{ and } \frac{d^2y}{dx^2} = 16x^2 (9 \log_e x)$$

$$\frac{dy}{dx} = 0 \rightarrow x = e^{\frac{1}{3}}$$

At $x = e^{\frac{1}{3}}, \frac{d^2y}{dx^2} > 0$, hence $x = e^{\frac{1}{3}}$ is point of minima

Also for $0 < x < 1, \frac{d^2y}{dx^2} < 0$ and for $x > 1, \frac{d^2y}{dx^2} > 0$

Hence $x = 1$ is point of inflection and for $x < 1$ graph is concave downward and for $x > 1$ graph is concave upward.

29. Which of the following statements is/are TRUE?

- (A) If f has domain $[0, \infty)$ and has no horizontal asymptotes then $\lim_{x \rightarrow \infty} f(x) = \infty$ or $\lim_{x \rightarrow \infty} f(x) = -\infty$
 (B) If f is continuous on $[-1, 1]$ and $f(-1) = 4$ and $f(1) = 3$ then there exist a number r such that $|r| < 1$ and $f(r) = \pi$
 (C) $\lim_{x \rightarrow \infty} \arcsin\left(\frac{x+1}{x}\right)$ does not exist
 (D) For all values of $m \in \mathbb{R}$ the line $y - mx + m - 1 = 0$ cuts the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ orthogonally

निम्न कथनों में सत्य कथन है :

- (A) यदि f का प्रान्त $[0, \infty)$ है और कोई भी क्षेतिज अनन्तस्पर्शी नहीं तो $\lim_{x \rightarrow \infty} f(x) = \infty$ या $\lim_{x \rightarrow \infty} f(x) = -\infty$
 (B) यदि $f, [-1, 1]$ में सतत् है तथा $f(-1) = 4$ और $f(1) = 3$ तो संख्या r विद्यमान होगी की $|r| < 1$ और $f(r) = \pi$
 (C) $\lim_{x \rightarrow \infty} \arcsin\left(\frac{x+1}{x}\right)$ विद्यमान नहीं है
 (D) रेखा $y - mx + m - 1 = 0$ वृत्त $x^2 + y^2 - 2x - 2y + 1 = 0$ को सभी $m \in \mathbb{R}$ के लिये लम्बवत् काटती है

Ans. BCD

Sol. (A) False, e.g. $f(x) = \sin \sqrt{x}$

(B) True, from IVT

(C) True as $\lim_{x \rightarrow \infty} \sin^{-1} \left(1 + \frac{1}{x} \right) = \sin^{-1}(\text{a quantity greater than one}) \Rightarrow \text{not defined}$

(D) True, as the line passes through the centre of the circle.

30. A function f is defined by $f(x) = \int_0^{\pi} \cos t \cos(x-t) dt$, $0 \leq x \leq 2\pi$ then which of the following hold(s) good :

(A) $f(x)$ is continuous but not differentiable in $(0, 2\pi)$

(B) Maximum value of f is π

(C) There exists atleast one $c \in (0, 2\pi)$ s.t. $f'(c) = 0$

(D) Minimum value of f is $-\frac{\pi}{2}$

यदि फलन $f(x) = \int_0^{\pi} \cos t \cos(x-t) dt$, $0 \leq x \leq 2\pi$ तो निम्न में से सत्य है :

(A) $f(x)$, $(0, 2\pi)$ में सतत् है लेकिन अवकलनीय नहीं

(B) $f(x)$ का अधिकतम मान π है

(C) कम से कम एक ऐसा बिंदुमान है कि $c \in (0, 2\pi)$ इस प्रकार कि $f'(c) = 0$.

(D) f का न्यूनतम मान $-\frac{\pi}{2}$ है

Ans. CD

Sol. $f(x) = \int_0^{\pi} \cos t \cos(x-t) dt$ (1)

$$= \int_0^{\pi} -\cos t \cdot \cos(x-\pi+t) dt \quad (\text{using King})$$

$$f(x) = \int_0^{\pi} \cos t \cdot \cos(x+t) dt \quad \dots(2)$$

(1) + (2) gives

$$2f(x) = \int_0^{\pi} \cos t (2 \cos x \cdot \cos t) dt$$

$$\therefore f(x) = \cos x \int_0^{\pi} \cos^2 t dt = 2 \cos x \int_0^{\pi/2} \cos^2 t dt$$

$$f(x) = \frac{\pi \cos x}{2} \text{ Now verify. Only (A) \& (B) are correct.}$$

Alternatively: Convert the integrand into sum of two cosine functions.

31. Let $f(x)$ be a polynomial of degree 3 having a local maximum at $x = -1$. If $f(-1) = 2$, $f(3) = 18$, and $f'(x)$ has a local minimum at $x = 0$, then :

(A) distance between $(-1, 2)$ and $(a, f(a))$, which are the points of local maximum and local minimum on the curve $y = f(x)$ is $2\sqrt{5}$

(B) $f(x)$ is a decreasing function for $1 \leq x \leq 2\sqrt{5}$

(C) $f'(x)$ has a local maximum at $x = 2\sqrt{5}$

(D) $f(x)$ has a local minimum at $x = 1$

दिया है $f(x)$ एक तीन घात का बहुपद फलन इस प्रकार से है कि जो $x = -1$ पर महत्तम है। यदि $f(-1) = 2$, $f(3) = 18$, तथा $f'(x)$ का $x = 0$ पर न्यूनतम है, तो :

(A) बिन्दुओं $(-1, 2)$ तथा $(a, f(a))$ के बीच की दूरी, जो कि वक्र $y = f(x)$ पर महत्तम एवं न्यूनतम बिन्दु है, $2\sqrt{5}$ है

(B) $f(x)$, $1 \leq x \leq 2\sqrt{5}$ के लिये ह्रासमान फलन है

(C) $f'(x)$, $x = 2\sqrt{5}$ पर उच्चिष्ठ है

(D) $f(x)$, $x = 1$ पर निम्निष्ठ है

Ans. AD

Sol. Let $f(x) = ax^3 + bx^2 + cx + d$

$$f'(x) = 3ax^2 + 2bx + c$$

$$f'(-1) = 0 \quad f(-1) = 2$$

$$3a - 2b + c = 0 \quad a + b - c + d = 2$$

$$f(3) = 18$$

$$27a + 90 + 3(1-d) = 18$$

$$f''(x) = 6ax + 2b$$

$$f''(0) = 0$$

$$b = 0$$

$$3a + c = 0$$

$$-a + b - c + d = 2$$

$$2a + d = 2$$

$$27a + 9b + 3c + d = 18$$

$$27a + 3c + d = 18$$

$$16a + 2 = 18$$

$$a = 1$$

$$c = -3$$

$$d = 0$$

$$f(x) = x^3 - 3x$$

32. Consider the function $f(x) = x^2 - x \sin x - \cos x$ then the statements which holds good, are :

(A) $f(x) = k$ has no solution for $k < -1$

(B) f is increasing for $x < 0$ and decreasing for $x > 0$

(C) $\lim_{x \rightarrow \pm\infty} f(x) \rightarrow \infty$

(D) The zeros of $f(x) = 0$ lie on the same side of the origin

माना फलन $f(x) = x^2 - x \sin x - \cos x$ तो निम्न में से सत्य कथन होंगे :

(A) $f(x) = k$ का कोई हल नहीं होगा यदि $k < -1$

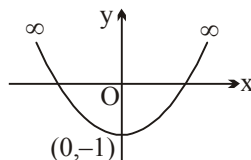
(B) f ; $x < 0$ के लिये वर्धमान तथा $x > 0$ के लिये ह्रसमान है

(C) $\lim_{x \rightarrow \pm\infty} f(x) \rightarrow \infty$

(D) $f(x) = 0$ के मूल बिन्दु के एक ही ओर होंगे

Ans. AC

Sol. Graph of $y = f(x) \Rightarrow$ (A) and (C)



33. For the function $f(x) = \ln(1 - \ln x)$ which of the following do not hold good?

(A) increasing in $(0, 1)$ and decreasing in $(1, e)$ (B) decreasing in $(0, 1)$ and increasing in $(1, e)$

(C) $x = 1$ is the critical number for $f(x)$. (D) f has two asymptotes

फलन $f(x) = \ln(1 - \ln x)$ के लिये असत्य है :

(A) $(0, 1)$ में वर्धमान और $(1, e)$ में ह्रसमान

(B) $(0, 1)$ में ह्रसमान और $(1, e)$ में वर्धमान

(C) $x = 1$, $f(x)$ का क्रान्तिक बिन्दु है

(D) f की दो अनन्त स्पर्शियाँ हैं

Ans. ABC

Sol. $f(x) = \ln(1 - \ln x)$

domain $(0, e)$

$$f'(x) = -\frac{1}{(1 - \ln x)} \cdot \frac{1}{x} < 0 \Rightarrow \text{decreasing } \forall x \text{ in its domain} \Rightarrow \text{(A) \& (B) are incorrect}$$

$$f'(1) = -1 \Rightarrow \text{(C) is also incorrect}$$

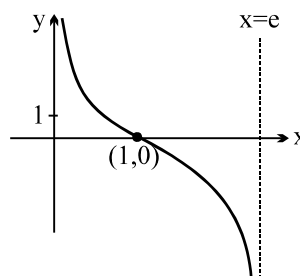
also $f(1) = 0$; $\lim_{x \rightarrow e^-} f(x) \rightarrow -\infty$; $\lim_{x \rightarrow 0^+} f(x) \rightarrow \infty$

$$f''(x) = \frac{-\ln x}{x^2(1 - \ln x)^2}$$

$f''(1) = 0$ which is a point of inflection

graph is as shown

y axis and $x = e$ are two asymptotes



34. Which of the following statement(s) is/are True :

- (A) If f is a continuous real valued function defined on $[0, 1]$ then $\exists$ positive number c such that $|f(x) - f(y)| \leq c$ for all x, y in $[0, 1]$.
- (B) If f is continuous on $\mathbb{R}$ such that $\lim_{x \rightarrow -\infty} f(x) = -\infty$ and $\lim_{x \rightarrow \infty} f(x) = \infty$ then the equation $f(x) = k$, $k \in \mathbb{R}$ has at least a root in $\mathbb{R}$
- (C) If f and g are derivable and $f(x) > g(x)$ for $a < x < b$ then $f'(x) > g'(x)$ for $a < x < b$
- (D) If f and g are continuous and $f(x) > g(x)$ for $a < x < b$ then $\int_a^b f(x) dx > \int_a^b g(x) dx$

निम्न में से सत्य कथन है :

- (A) यदि f वास्तविक मानों का सतत् फलन $[0, 1]$ में परिभाषित है तो धनात्मक c इस प्रकार से होगा की $|f(x) - f(y)| \leq c$, $\forall x, y \in [0, 1]$.
- (B) यदि $f, \mathbb{R}$ में सतत् है इस प्रकार की $\lim_{x \rightarrow -\infty} f(x) = -\infty$ और $\lim_{x \rightarrow \infty} f(x) = \infty$ तो $f(x) = k$, $k \in \mathbb{R}$ का कम से कम एक मूल $\mathbb{R}$ में है
- (C) यदि f और g अवकलनीय है और $f(x) > g(x)$ for $a < x < b$ के लिये तो $f'(x) > g'(x)$; $a < x < b$
- (D) यदि f और g सतत् फलन है और $f(x) > g(x)$; $a < x < b$ तो $\int_a^b f(x) dx > \int_a^b g(x) dx$

Ans. ABD

Sol. (A) as f is continuous, f is bounded on a closed interval $\Rightarrow$ True

(B) Consider $g(x) = f(x) - k$ obviously g is continuous and $\lim_{x \rightarrow \infty} g(x) = \infty$ and $\lim_{x \rightarrow -\infty} g(x) = -\infty$ hence $g(x)$ must vanish at least once in $(-\infty, \infty)$

$\therefore f(x) - k = 0 \Rightarrow f(x) = k$ (using IVT)

(D) Consider $f(x) = \cos x$ and $g(x) = \sin x$ on $\left(0, \frac{\pi}{4}\right)$.

Note that $f(x) > g(x)$ on $\left(0, \frac{\pi}{4}\right)$

now $f'(x) = -\sin x$ and $g'(x) = \cos x$

but $f'(x) \not> g'(x)$ on $\left(0, \frac{\pi}{4}\right)$ i.e. $-\sin x \not> \cos x$ in $\left(0, \frac{\pi}{4}\right)$